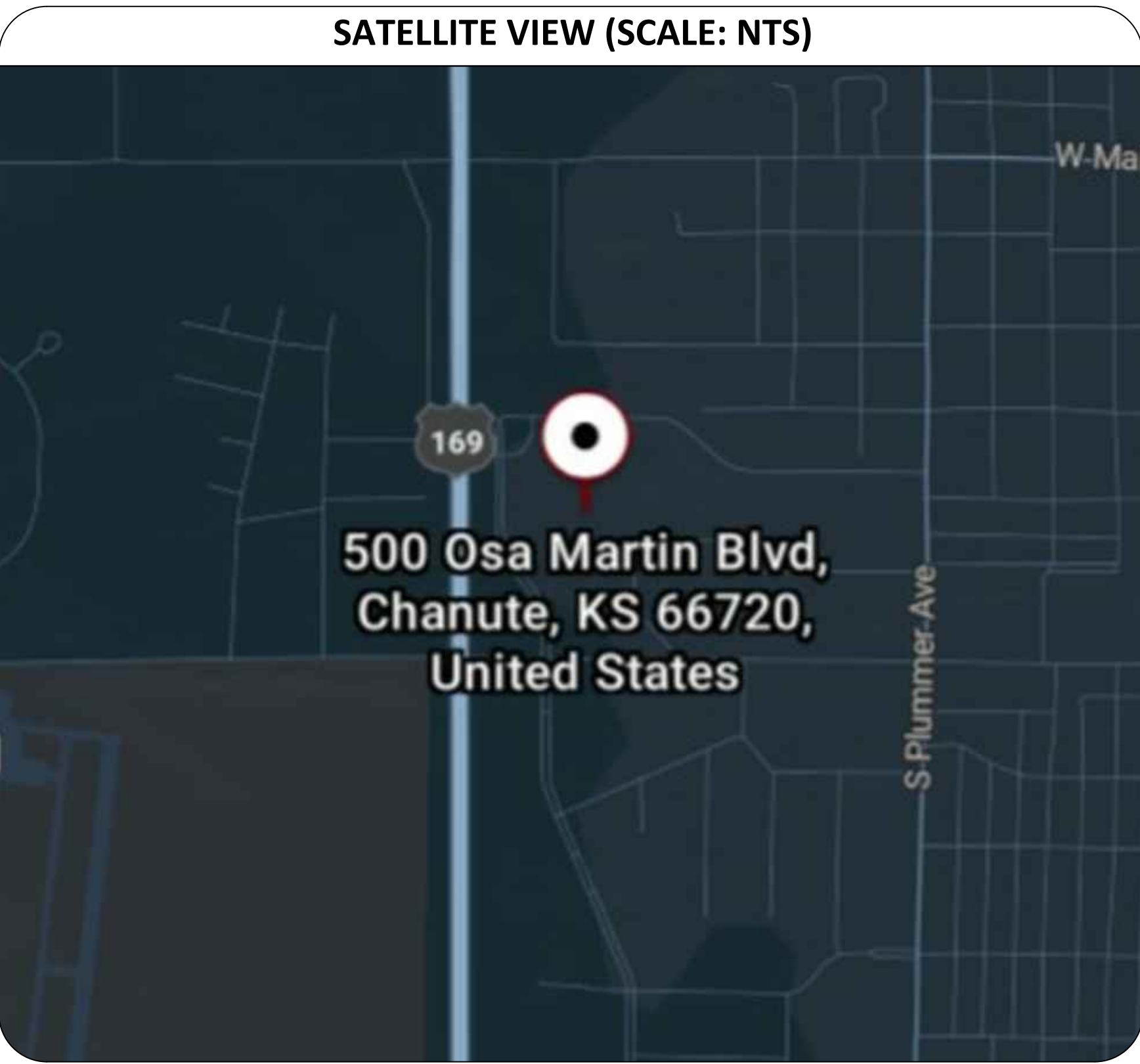


PROJECT: PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(1208) MODULES x 600W	(5) INVERTER x (100000W)
724.800 kW	500.000 kW



SCOPE OF WORK
- PROPOSED PROJECT TO INSTALL GRID- TIED PV SYSTEM ON AN EXISTING ROOFTOP STRUCTURE AT A COMMERCIAL LOCATION: 1234 STREET NAME, CHANUTE, KANSAS 9874. COORDINATES: 37.676976, -95.477182 - THE PV MODULES WILL BE SECURED ON ROOF USING PRE-ENGINEERED ROOF RACKING SYSTEM. - INSTALLED PV SYSTEM WILL BE CONNECTED TO BUILDING SERVICE EQUIPMENT USING APPLICABLE INTERCONNECTION METHOD (AS PER AHJ). - THE ENERGY PRODUCED BY PV SYSTEM WILL BE USED TO POWER THE ENERGY REQUIREMENTS. EXCESS POWER IS SENT BACK TO THE UTILITY GRID.

APPLICABLE GOVERNING CODES & NOTES
THIS PROJECT SHALL COMPLY WITH THE FOLLOWING CODES:
2006 INTERNATIONAL BUILDING CODE 2006 INTERNATIONAL FIRE CODE 2020 NFPA70, NATIONAL ELECTRICAL CODE
ALL OTHER RELATED STATE & AHJ BY-LAWS & ORDINANCES

DESIGN CRITERIA
- CONSTRUCTION: COMMERCIAL - EXPOSURE CATEGORY : C, RISK CATEGORY : III - WIND SPEED (ASCE 7-10) : 109 MPH - GROUND SNOW LOAD (ASCE 7-10) : 24 lb/sqft - ASHRAE EXTREME LOW: -18°C - ASHRAE HIGH TEMP (2% AVG.) : 35°C - ASHRAE DISTANCE ABOVE ROOF (7/8") : 57°C

SHEET INDEX	
PV-0	PROJECT SUMMARY
PV-1	SITE PLAN
PV-2	ROOF ATTACHMENT DETAILS
PV-3	BOM & STRING LAYOUT
S-1	STRUCTURAL DETAILED DIAGRAM
S-2	EQUIPMENT ELEVATION DIAGRAM
E-1	ELECTRICAL DIAGRAM & CALCULATIONS
E-2	PLACARDS
E.S	EQUIPMENT SPECIFICATIONS

EQUIPMENT DETAILS	
(#) PV MODULES	(1208) PHILADELPHIA SOLAR PS-MNB144(HCBF)-600W
(#) RAPID SHUTDOWN SWITCH	(1208) AP SMART RSD-S-PLC
(#) INVERTER	(5) SOLIS S5-GC100K-US INVERTER
INVERTER CURRENT	601.50A/(480V)
ROOF RACKING SYSTEM	S5! S-5-PVKIT-2.0 & SOLARSTACK SS8-TK10
STRINGS	38 STRINGS OF 18 MODULES, 8 STRINGS OF 17 MODULES, 8 STRINGS OF 16 MODULES, 6 STRINGS OF 15 MODULES, 4 STRINGS OF 14 MODULES, 3 STRINGS OF 13 MODULES, 2 STRINGS OF 12 MODULES, 1 STRING OF 11 MODULES, 4 STRINGS OF 10 MODULES
UTILITY AHJ	CITY OF CHANUTE

GENERAL NOTES
- ALL COMPONENTS ARE UL LISTED & NEC CERTIFIED, WHERE WARRANTED - A DISCONNECTING MEANS SHALL BE INSTALLED AT A READILY ACCESSIBLE LOCATION EITHER OUTSIDE OF THE BUILDING OR STRUCTURE OR INSIDE NEAREST POINT OF ENTRANCE OF THE SYSTEM CONDUCTORS. - HEIGHT OF THE INTEGRATED AC/DC DISCONNECT SHALL NOT EXCEED 6'-7" PER NEC CODE 240.24. - A GROUNDING ELECTRODE SYSTEM IN ACCORDANCE WITH NEC 690.47 AND 250.50 THROUGH 60 AND 250-166 SHALL BE PROVIDED. PER NEC GROUNDING ELECTRODE SYSTEM OF EXISTING BUILDING MAY BE USED AND BONDED TO THE SERVICE ENTRANCE. IF EXISTING SYSTEM IS INACCESSIBLE OR INADEQUATE A SUPPLEMENTAL GROUNDING ELECTRODE WILL BE USED AT THE INVERTER LOCATION CONSISTING OF A UL LISTED 8 FT. GROUND ROD WITH ACORN CLAMP. GROUNDING ELECTRODE CONDUCTORS SHALL BE NO LESS THAN #8 AWG AND NO LARGER THAN #6 AWG COPPER AND BONDED TO THE EXISTING GROUNDING ELECTRODE TO PROVIDE FOR A COMPLETE SYSTEM. - PHOTOVOLTAIC MODULES ARE TO BE CONSIDERED NON-COMBUSTIBLE. - PHOTOVOLTAIC INSTALLATION WILL NOT OBSTRUCT ANY PLUMBING, MECHANICAL, OR BUILDING ROOF VENTS. - WATER PROOF CONNECTORS AND HUBS SHALL BE USED WHERE APPLICABLE PER NEC 312.2 AND 314.15. - GROUNDING/BONDING BUSHINGS SHALL BE INSTALLED WHERE APPLICABLE PER NEC 250.92. - ALL EXTERIOR RACEWAYS ON WALLS SHOULD BE EMT UNLESS NOTED OTHERWISE. - CONTRACTOR SHALL INSTALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS & NOTHING IN THIS DOCUMENT SHALL BE INTERPRETED IN A WAY THAT OVERRIDES THEM. - CONTRACTOR IS RESPONSIBLE FOR VERIFICATION OF ALL CONDITIONS, DIMENSIONS & DETAILS IN THIS DOCUMENT. ALL DIMENSIONS SHALL BE VERIFIED BY SUBCONTRACTOR BEFORE INSTALLATION.
PHOTOVOLTAIC SYSTEM FIRE CLASSIFICATION LISTING IN ACCORDANCE WITH UL1703 STANDARD.

ELECTRICAL NOTES
- THE EQUIPMENT AND ALL ASSOCIATED WIRING AND INTERCONNECTION SHALL BE INSTALLED ONLY BY QUALIFIED PEOPLE. A QUALIFIED PERSON IS ONE WHO HAS SKILLS AND KNOWLEDGE RELATED TO THE CONSTRUCTION AND OPERATION OF THE ELECTRICAL EQUIPMENT AND INSTALLATIONS AND HAS RECEIVED SAFETY TRAINING TO RECOGNIZE AND AVOID THE HAZARDS INVOLVED. (NEC690.4(E) AND 705.6) - LOCAL UTILITY PROVIDER SHALL BE NOTIFIED PRIOR TO USE AND ACTIVATION OF ANY SOLAR PHOTOVOLTAIC INSTALLATION. FOR A LINE SIDE TAP CONNECTION, UTILITY NEEDS TO BE NOTIFIED WELL IN ADVANCE TO COORDINATE BUILDING ELECTRICAL SHUT OFF. - NEW CONDUIT ROUTING SHOWN IS ESSENTIALLY SCHEMATIC. SUBCONTRACTOR SHALL LAY OUT RUNS TO SUIT FIELD CONDITIONS AND THE COORDINATION REQUIREMENTS OF OTHER TRADES. - ARRAY WIRING SHOULD NOT BE READILY ACCESSIBLE EXCEPT TO QUALIFIED PERSONNEL. - ALL EXTERIOR CONDUIT, FITTINGS, AND BOXES SHALL BE WATERTIGHT AND APPROVED FOR USE IN WET LOCATIONS . - WIRING METHODS FOR PV SYSTEM CONDUCTORS AREN'T PERMITTED WITHIN 10 IN. OF THE ROOF DECKING OR SHEATHING EXCEPT WHERE LOCATED DIRECTLY BELOW THE ROOF SURFACE THAT'S COVERED BY PV MODULES AND ASSOCIATED EQUIPMENT WIRING - BACK-FED BREAKER MUST BE AT THE OPPOSITE END OF BUS BAR FROM THE MAIN BREAKER OR MAIN LUG SUPPLYING CURRENT FROM THE UTILITIES. - ALL CONDUCTORS AND WIRE TIES EXPOSED TO SUNLIGHT ARE LISTED AS UV RESISTANT. - CONTRACTOR SHALL FOLLOW ALL ELECTRICAL EQUIPMENT LABELING REQUIREMENTS ACCORDING TO NEC. - MEASURE THE LINE-TO-LINE AND LINE-TO-NEUTRAL VOLTAGE OF ALL SERVICE ENTRANCE CONDUCTORS PRIOR TO INSTALLING ANY SOLAR EQUIPMENT.

CONTRACTOR INFO

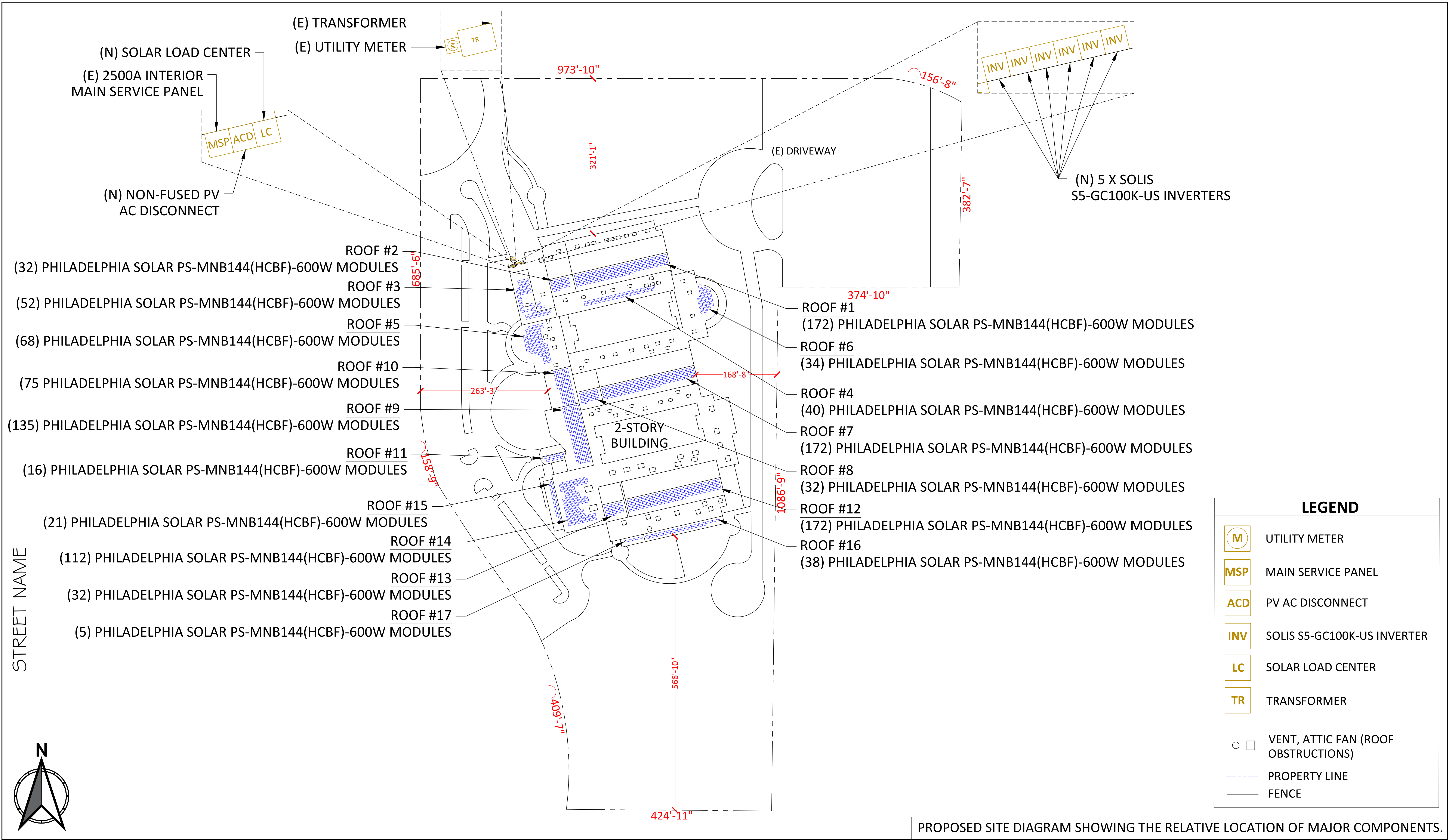
DATE	11/27/2025
PROJECT ID	SP-1
CREATED BY	TV
CHECKED BY	MSB
SCALE	NTS
SHEET SIZE	24"X 36"
	ANSI D

COMMERCIAL
1234 STREET NAME, CHANUTE, KANSAS 9874
APN: 147852369789456123
BUILDING OWNER INFO

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

PV-0
SHEET NAME
PROJECT SUMMARY
SHEET INFORMATION

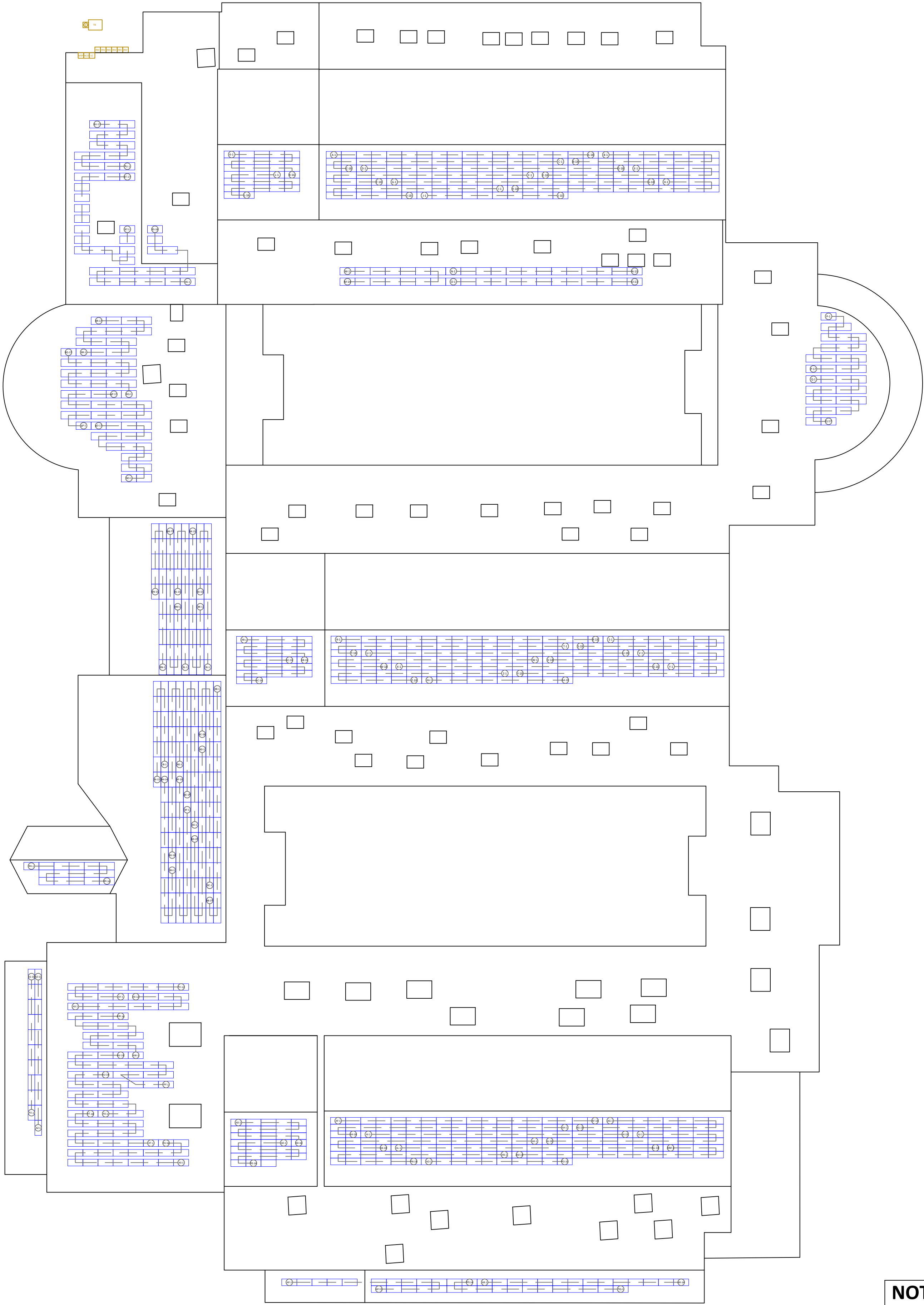
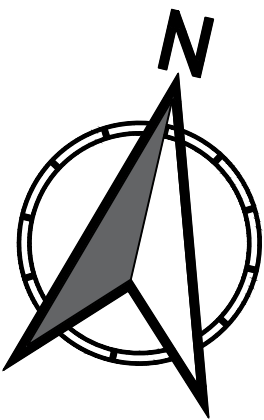


PROPOSED SITE DIAGRAM SHOWING THE RELATIVE LOCATION OF MAJOR COMPONENTS.

CONTRACTOR INFO	<table><tr><td>DATE</td><td>11/27/2025</td></tr><tr><td>PROJECT ID</td><td>SP-1</td></tr><tr><td>CREATED BY</td><td>TV</td></tr><tr><td>CHECKED BY</td><td>MSB</td></tr><tr><td>SCALE</td><td>1/85" = 1'-0"</td></tr><tr><td>SHEET SIZE</td><td>24"X 36"</td></tr><tr><td></td><td>ANSI D</td></tr></table>	DATE	11/27/2025	PROJECT ID	SP-1	CREATED BY	TV	CHECKED BY	MSB	SCALE	1/85" = 1'-0"	SHEET SIZE	24"X 36"		ANSI D	BUILDING OWNER INFO	COMMERCIAL 1234 STREET NAME, CHANUTE, KANSAS 9874 APN: 147852369789456123	<table><tr><th>#</th><th>REVISION/UPDATE</th><th>DATE</th></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr></table>	#	REVISION/UPDATE	DATE																SIGNED & STAMP BY	PV-1 SHEET NAME SITE PLAN SHEET INFORMATION
	DATE	11/27/2025																																				
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SHEET SIZE	24"X 36"																																					
	ANSI D																																					
#	REVISION/UPDATE	DATE																																				

STRING LEGEND

A	STRING #1- 18 MODULES	Z	STRING #26- 18 MODULES	AY	STRING #51- 12 MODULES
B	STRING #2- 18 MODULES	AA	STRING #27- 10 MODULES	AZ	STRING #52- 11 MODULES
C	STRING #3- 18 MODULES	AB	STRING #28- 16 MODULES	A1	STRING #53- 10 MODULES
D	STRING #4- 18 MODULES	AC	STRING #29- 16 MODULES	BA	STRING #53- 16 MODULES
E	STRING #5- 18 MODULES	AD	STRING #30- 16 MODULES	BB	STRING #54- 18 MODULES
F	STRING #6- 18 MODULES	AE	STRING #31- 16 MODULES	BC	STRING #55- 18 MODULES
G	STRING #7- 18 MODULES	AF	STRING #32- 18 MODULES	BD	STRING #56- 18 MODULES
H	STRING #8- 18 MODULES	AG	STRING #33- 18 MODULES	BE	STRING #57- 18 MODULES
I	STRING #9- 18 MODULES	AH	STRING #34- 18 MODULES	BF	STRING #58- 18 MODULES
J	STRING #10- 10 MODULES	AI	STRING #35- 18 MODULES	BG	STRING #59- 18 MODULES
K	STRING #11- 16 MODULES	AJ	STRING #36- 18 MODULES	BH	STRING #60- 14 MODULES
L	STRING #12- 16 MODULES	AK	STRING #37- 18 MODULES	BI	STRING #61- 13 MODULES
M	STRING #13- 14 MODULES	AL	STRING #38- 18 MODULES	BJ	STRING #62- 15 MODULES
N	STRING #14- 13 MODULES	AM	STRING #39- 18 MODULES	BK	STRING #63- 15 MODULES
O	STRING #15- 13 MODULES	AN	STRING #40- 18 MODULES	BL	STRING #64- 15 MODULES
P	STRING #16- 17 MODULES	AO	STRING #41- 10 MODULES	BM	STRING #65- 15 MODULES
Q	STRING #17- 17 MODULES	AP	STRING #42- 15 MODULES	BN	STRING #66- 15 MODULES
R	STRING #18- 18 MODULES	AQ	STRING #43- 14 MODULES	BO	STRING #67- 17 MODULES
S	STRING #19- 18 MODULES	AR	STRING #44- 14 MODULES	BP	STRING #68- 17 MODULES
T	STRING #20- 18 MODULES	AS	STRING #45- 18 MODULES	BQ	STRING #69- 17 MODULES
U	STRING #21- 18 MODULES	AT	STRING #46- 18 MODULES	BR	STRING #70- 17 MODULES
V	STRING #22- 18 MODULES	AU	STRING #47- 18 MODULES	BS	STRING #71- 18 MODULES
W	STRING #23- 18 MODULES	AV	STRING #48- 18 MODULES	BT	STRING #72- 17 MODULES
X	STRING #24- 18 MODULES	AW	STRING #49- 16 MODULES	BU	STRING #73- 17 MODULES
Y	STRING #25- 18 MODULES	AX	STRING #50- 12 MODULES		



LEGEND

M	UTILITY METER
MSP	MAIN SERVICE PANEL
ACD	PV AC DISCONNECT
INV	SOLIS S5-GC100K-US INVERTER
TR	TRANSFORMER
○ □	VENT, ATTIC FAN (ROOF OBSTRUCTIONS)
----	CONDUIT

NOTES: BATTERY SHOULD BE INSTALLED 3FEET DISTANCE FROM ANY ENTERING DWELLING UNIT.

CONTRACTOR INFO

DATE	11/27/2025
PROJECT ID	SP-1
CREATED BY	TV
CHECKED BY	MSB
SCALE	1/34" = 1'-0"
SHEET SIZE	24"X 36"
	ANSI D

BUILDING OWNER INFO

COMMERCIAL
1234 STREET NAME, CHANUTE, KANSAS 9874
APN: 147852369789456123

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

PV-3

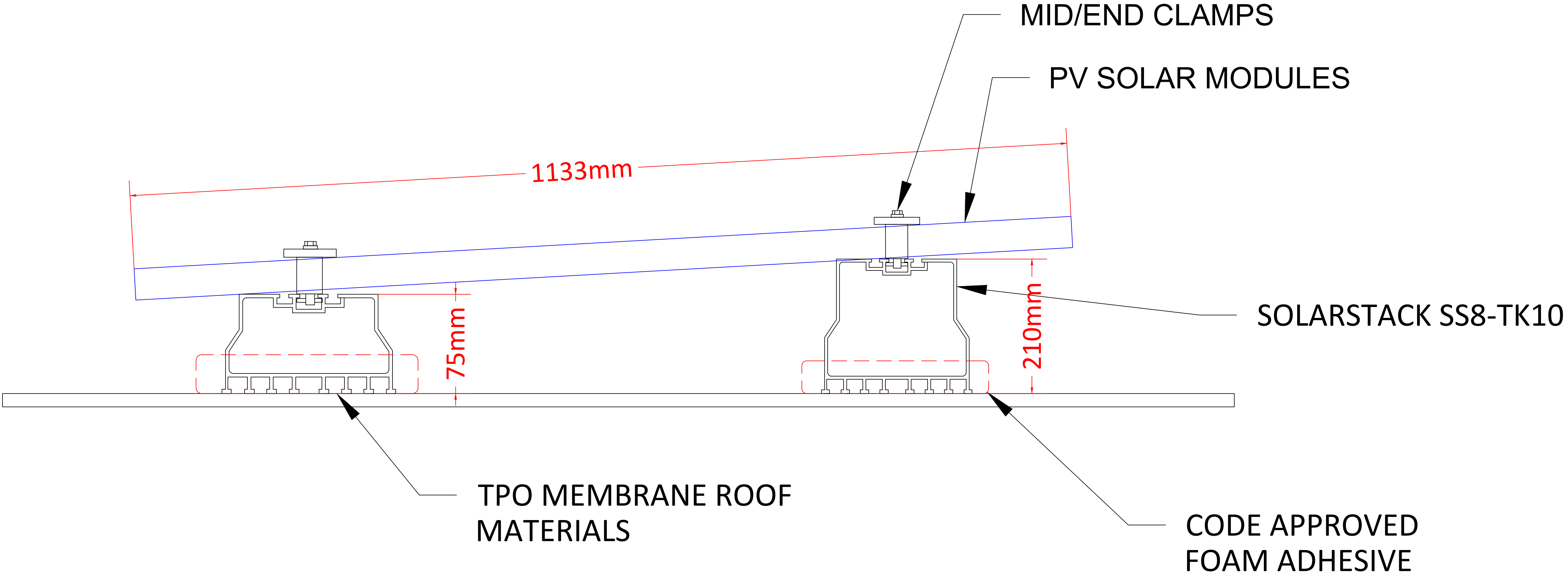
SHEET NAME
BOM &
STRING
LAYOUT

SHEET INFORMATION

STRUCTURAL NOTES:

THE SOLAR PANELS ARE TO BE MOUNTED TO THE ROOF FRAMING USING THE SYSTEM WITH SOLARSTACK SS8-TK10

- 1. ATTACHMENTS.
- 2. THE MOUNTING FEET ARE TO BE SPACED AS SHOWN IN THE DETAILS, AND MUST BE STAGGERED TO ADJACENT FRAMING MEMBERS TO SPREAD OUT THE ADDITIONAL LOAD.
- 3. ROOF LIVE LOAD = 20 PSF
- 4. GROUND SNOW LOAD = 24 lb/sqft
- 5. WIND SPEED = 109 MPH
- 6. EXPOSURE CATEGORY = C
- 7. RISK CATEGORY = III



NOTE: - CONTRACTOR/INSTALLER TO VERIFY COMPATIBILITY OF ANY BRANDS OR PRODUCTS SUBSTITUTED OR USED AS ALTERNATES WITHIN ANY BRAND-SPECIFIC SYSTEMS. CONTRACTOR SHALL SUPPLY AND PRESENT CERTIFICATES OF COMPATIBILITY TO THE BUILDING OFFICIAL UPON INSPECTION.

CONTRACTOR INFO	DATE	11/27/2025
	PROJECT ID	SP-1
	CREATED BY	TV
	CHECKED BY	MSB
	SCALE	NTS
SHEET SIZE	24"X 36"	
	ANSI D	

BUILDING OWNER INFO	COMMERCIAL	
	1234 STREET NAME, CHANUTE, KANSAS 9874	
	APN: 147852369789456123	

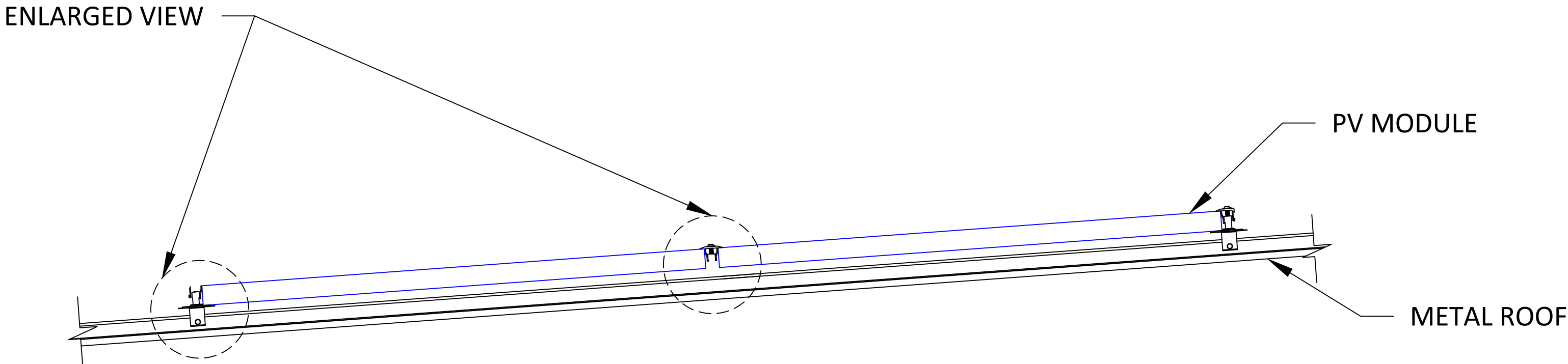
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SIGNED & STAMP BY	

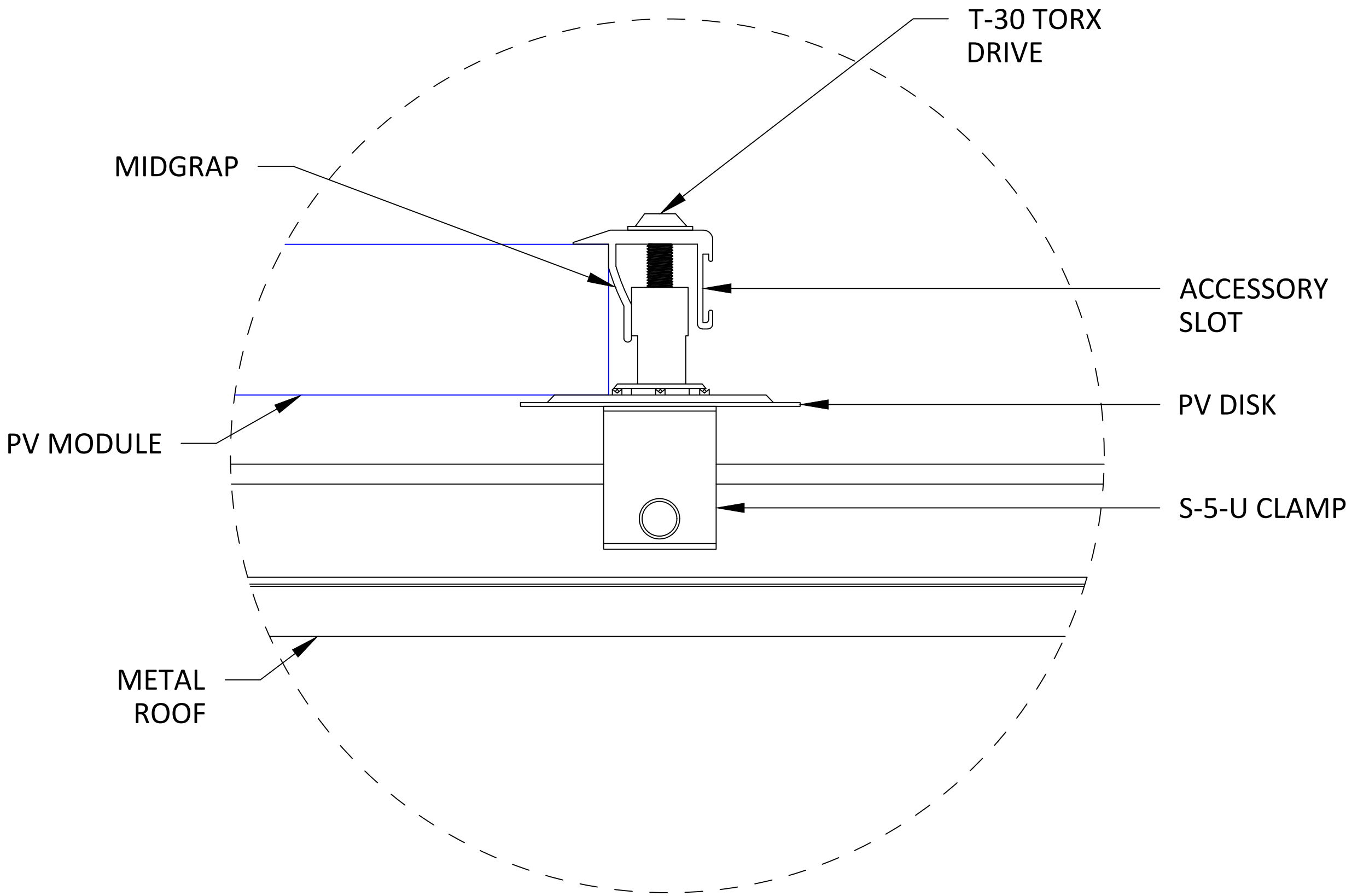
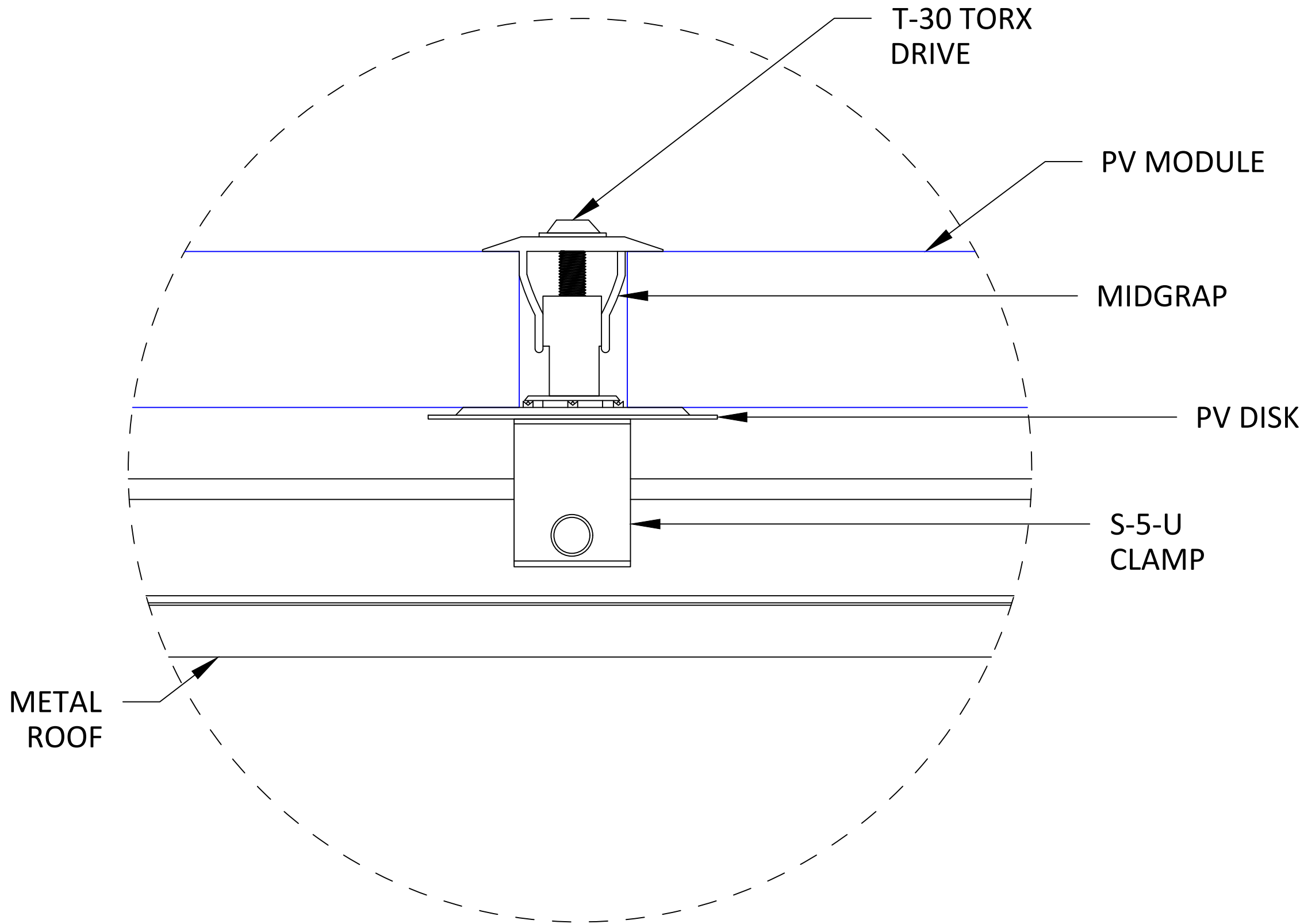
S-1
SHEET NAME
STRUCTURAL DETAILED DIAGRAM
SHEET INFORMATION

STRUCTURAL NOTES:

- 1. THE SOLAR PANELS ARE TO BE MOUNTED TO THE ROOF FRAMING USING THE SYSTEM WITH S5! S-5-PVKIT-2.0 ATTACHMENTS.
- 2. THE MOUNTING FEET ARE TO BE SPACED AS SHOWN IN THE DETAILS, AND MUST BE STAGGERED TO ADJACENT FRAMING MEMBERS TO SPREAD OUT THE ADDITIONAL LOAD.
- 3. ROOF LIVE LOAD = 20 PSF
- 4. GROUND SNOW LOAD = 24 lb/sqft
- 5. WIND SPEED = 109 MPH
- 6. EXPOSURE CATEGORY = C
- 7. RISK CATEGORY = III



STRUCTURAL ELEVATION DIAGRAM



ATTACHMENT DETAIL (SIDE VIEW)

NOTE: - CONTRACTOR/INSTALLER TO VERIFY COMPATIBILITY OF ANY BRANDS OR PRODUCTS SUBSTITUTED OR USED AS ALTERNATES WITHIN ANY BRAND-SPECIFIC SYSTEMS. CONTRACTOR SHALL SUPPLY AND PRESENT CERTIFICATES OF COMPATIBILITY TO THE BUILDING OFFICIAL UPON INSPECTION.

CONTRACTOR INFO

DATE	11/27/2025
PROJECT ID	SP-1
CREATED BY	TV
CHECKED BY	MSB
SCALE	NTS
SHEET SIZE	24"X 36"
	ANSI D

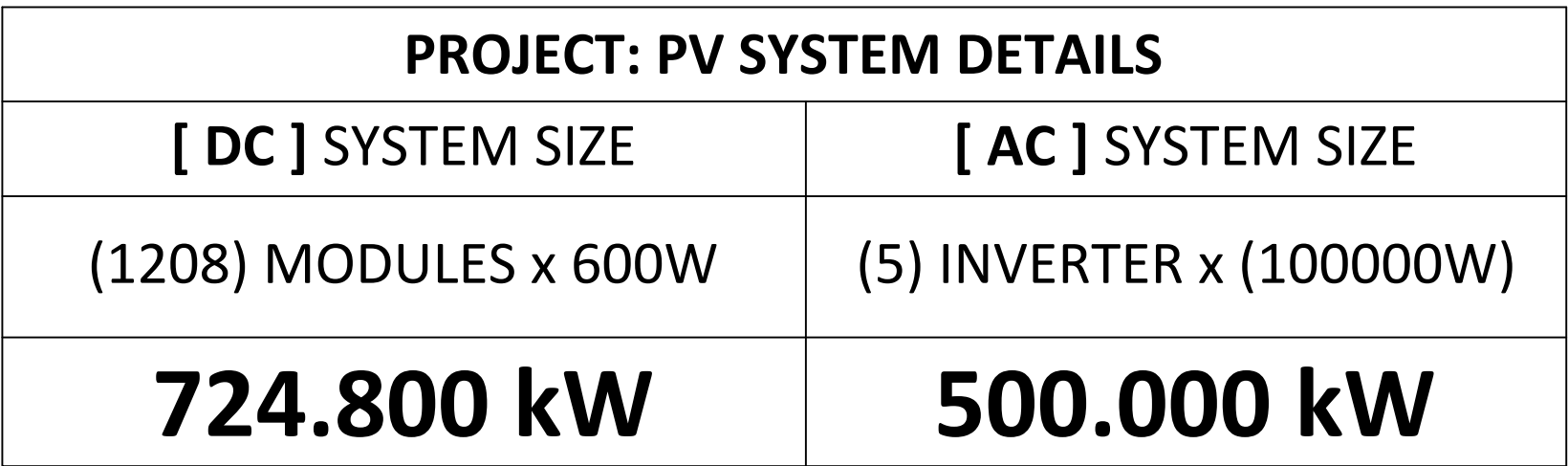
BUILDING OWNER INFO

COMMERCIAL
1234 STREET NAME, CHANUTE, KANSAS 9874
APN: 147852369789456123

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

S-2
SHEET NAME
EQUIPMENT
ELEVATION
DIAGRAM
SHEET INFORMATION



AMBIENT TEMPERATURE SPECS	
RECORD LOW TEMP	-19°C
AMBIENT TEMP (HIGH TEMP 2%)	35°C
CONDUIT HEIGHT	7/8"
ROOF TOP TEMP	90°C
CONDUCTOR TEMPERATURE RATE	57°C
MODULE TEMPERATURE COEFFICIENT OF VOC	-0.25 %/°C

EQUIPMENT DETAILS	
(#) PV MODULES	(1208) PHILADELPHIA SOLAR PS-MNB144(HCBF)-600W
(#) RAPID SHUTDOWN SWITCH	(1208) AP SMART RSD-S-PLC
(#) INVERTER	(5) SOLIS S5-GC100K-US INVERTERS
INVERTER CURRENT	601.50A/(480V)
STRINGS	38 STRINGS OF 18 MODULES, 8 STRINGS OF 17 MODULES, 8 STRINGS OF 16 MODULES, 6 STRINGS OF 15 MODULES, 4 STRINGS OF 14 MODULES, 3 STRINGS OF 13 MODULES, 2 STRINGS OF 12 MODULES, 1 STRING OF 11 MODULES, 4 STRINGS OF 10 MODULES
UTILITY	CITY OF CHANUTE
AHJ	CITY OF CHANUTE

E-1 SHEET NAME ELECTRICAL DIAGRAM & CALCULATIONS SHEET INFORMATION
--

ID	CONDUCTOR			CONDUIT	# OF PARALLEL CIRCUITS	CURRENT-CARRYING CONDUCTORS IN CONDUIT	CONDUIT FILL PERCENT	OCPD	EGC		TEMP. CORR. FACTOR		CONDUIT FILL FACTOR	CONT. CURRENT (IQ8PLUS CONTINUOUS CURRENT x NO. OF IQ8PLUS)				MAX CURRENT	BASE AMP.	DERATED AMP.	LENGTH	VOLTAGE DROP
A	10 AWG	PV WIRE		-	1	2	N/A	N/A	6 AWG	BARE COPPER	0.71	(57°C)	N/A	13.4A	x	1.25	=	16.8A	N/A	N/A	73FT	1.18%
B	2/0 AWG	THWN-2	COPPER	MIN 2" Dia PVC	1	4	29.02%	175A	2 AWG	THWN-2, COPPER	0.96	(35°C)	0.8	120.3A	x	1.25	=	150.4A	195A	195A x 0.8 x 0.96 =149.7A	26FT	0.84%
C	400 kcmil	THWN-2	COPPER	MIN 5" Dia PVC	2	8	27.83%	800A	2 AWG	THWN-2, COPPER	0.96	(35°C)	1	300.8A	x	1.25	=	375.9A	430A	430A x1x 0.96 = 412.8A	10FT	0.03%
D	400 kcmil	THWN-2	COPPER	MIN 5" Dia PVC	2	8	27.26%	N/A	-	-	0.96	(35°C)	1	300.8A	x	1.25	=	375.9A	430A	430A x1x 0.96 = 412.8A	5FT	0.01%

CONTRACTOR INFO	DATE	11/27/2025	COMMERCIAL 1234 STREET NAME, CHANUTE, KANSAS 9874 APN: 147852369789456123	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	E-1.1 SHEET NAME ELECTRICAL DIAGRAM & CALCULATIONS SHEET INFORMATION
	PROJECT ID	SP-1						
	CREATED BY	TV						
	CHECKED BY	MSB						
	SCALE	NTS						
	SHEET SIZE	24"X 36"						
	ANSI D	BUILDING OWNER INFO						

⚠️

WARNING

ELECTRICAL SHOCK HAZARD

TERMINALS ON LINE AND LOAD SIDES MAY BE ENERGIZED IN THE OPEN POSITION

LABEL LOCATION:
INVERTER(S), AC DISCONNECT(S), AC COMBINER PANEL (IF APPLICABLE).
PER CODE(S): NEC 2020: NEC 706.15 (C)(4) & NEC 690.13(B)

RAPID SHUTDOWN FOR SOLAR PV SYSTEM

LABEL LOCATION:
UTILITY SERVICE ENTRANCE/METER, INVERTER/DC DISCONNECT IF REQUIRED BY LOCAL AHJ, OR OTHER LOCATIONS AS REQUIRED BY LOCAL AHJ.
PER CODE(S): NEC 2020: 690.56(C)(2)

⚠️

WARNING

POWER SOURCE OUTPUT CONNECTION DO NOT RELOCATE THIS OVERCURRENT DEVICE

LABEL LOCATION:
SERVICE PANEL IF SUM OF BREAKERS EXCEEDS PANEL RATING
NEC 705.12 (B)(3)(2)

⚠️

WARNING

DUAL POWER SOURCE
SECOND SOURCE IS PHOTOVOLTAIC SYSTEM

LABEL LOCATION:
POINT OF INTERCONNECTION
PRODUCTION METER
NEC 705.12(B)(3)(2) & NEC 690.59)

PHOTOVOLTAIC AC DISCONNECT

MAXIMUM AC OPERATING CURRENT: 174.5A AMPS
NOMINAL OPERATING AC VOLTAGE: 480 VAC

LABEL LOCATION:
AC DISCONNECT(S), PHOTOVOLTAIC SYSTEM POINT OF INTERCONNECTION.
PER CODE(S): NEC 2020: 690.54

PHOTOVOLTAIC POWER SOURCE

LABEL LOCATION:
CONDUIT, COMBINER BOX
(PER CODE: NEC690.31(D)(2)

⚠️

CAUTION

PHOTOVOLTAIC SYSTEM CIRCUIT IS BACKFED

LABEL LOCATION:
MSP (PER CODE: NEC 705.12(D) & NEC 690.59

MAIN PHOTOVOLTAIC SYSTEM DISCONNECT

LABEL LOCATION:
MAIN SERVICE DISCONNECT / UTILITY METER
(PER CODE: NEC 690.13(B))

⚠️

WARNING

THIS EQUIPMENT FED BY MULTIPLE SOURCES:
TOTAL RATING OF ALL OVERCURRENT DEVICES EXCLUDING MAIN POWER SUPPLY SHALL NOT EXCEED AMPACITY OF BUSBAR

LABEL LOCATION:
POINTS OF CONNECTION/BREAKER
CODE: NEC 705.12(B)(3)(2)

⚠️

WARNING

ELECTRIC SHOCK HAZARD

TERMINALS ON BOTH LINE AND LOAD SIDES MAY BE ENERGIZED IN THE OPEN POSITION

DC VOLTAGE IS ALWAYS PRESENT WHEN SOLAR MODULES ARE EXPOSED TO SUNLIGHT

LABEL LOCATION:
DC DISCONNECT, POINT OF INTERCONNECTION
(PER CODE: NEC 690.13(B))

⚠️

WARNING

THE DISCONNECTION OF THE GROUNDED CONDUCTOR(S) MAY RESULT IN OVERVOLTAGE ON THE EQUIPMENT

LABEL LOCATION:
INVERTER
PER CODE: NEC 690.31(E)

PHOTOVOLTAIC AC DISCONNECT

LABEL LOCATION:
AC DISCONNECT/BREAKER/ POINT OF CONNECTION
(PER CODE: NEC 690.13(B)

PV SOLAR BREAKER

DO NOT RELOCATE THIS OVERCURRENT DEVICE

LABEL LOCATION:
MAIN SERVICE PANEL
2022 CEC 705.12(B)(3)(2)

7"

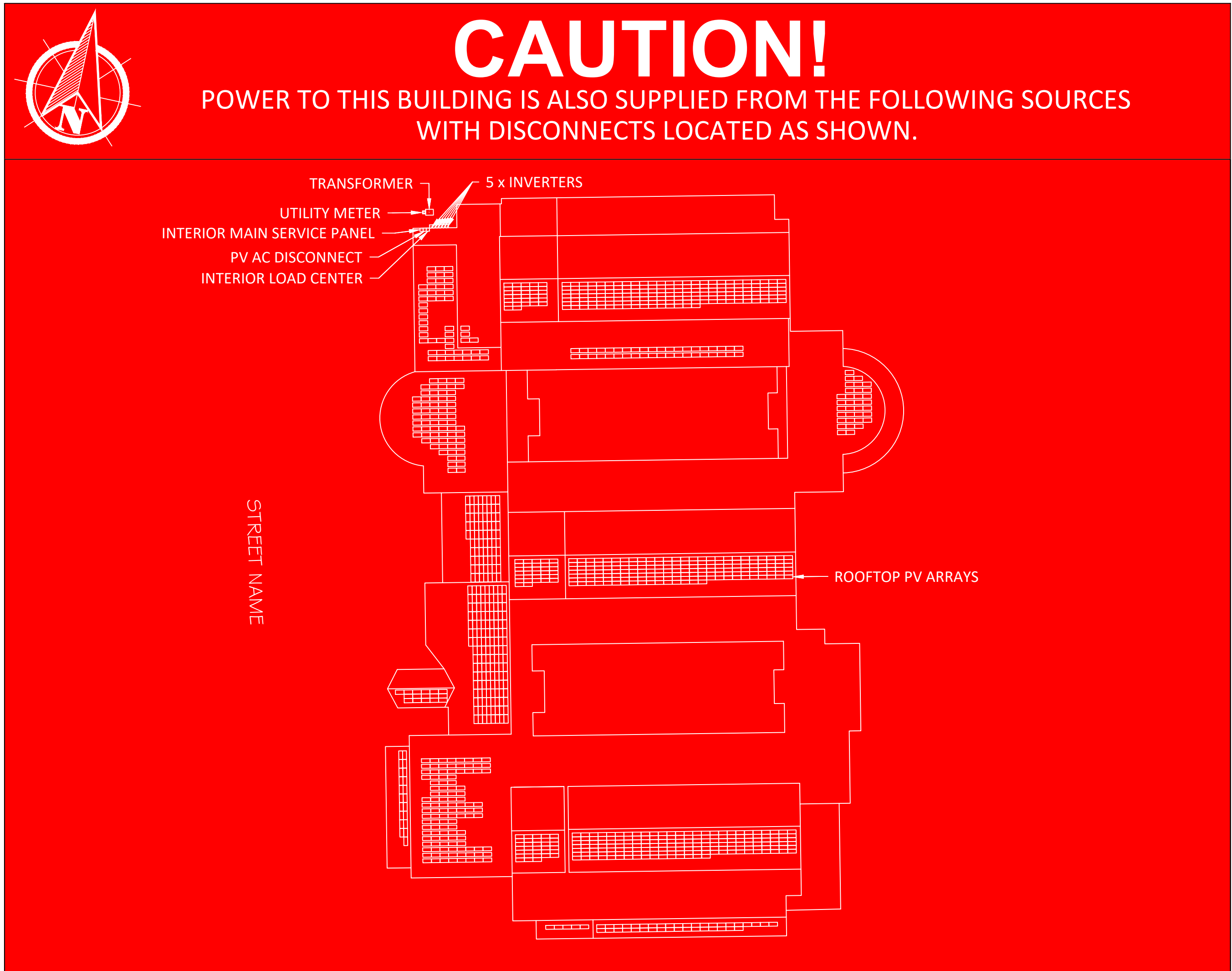
6"

SOLAR PV SYSTEM EQUIPPED WITH RAPID SHUTDOWN

TURN RAPID SHUTDOWN SWITCH TO THE "OFF" POSITION TO SHUT DOWN PV SYSTEM AND REDUCE SHOCK HAZARD IN THE ARRAY.

SOLAR ELECTRIC PV PANELS

LABEL LOCATION:
ON OR NO MORE THAT 1 M (3 FT) FROM THE SERVICE DISCONNECTING MEANS TO WHICH THE PV SYSTEMS ARE CONNECTED.
PER CODE(S): NEC 2020: IFC 690.56(C)



LABEL LOCATION:
EACH SERVICE EQUIPMENT LOCATION AND AT THE LOCATION(S) OF THE SYSTEM DISCONNECT(S) FOR ALL ELECTRIC POWER PRODUCTION SOURCES CAPABLE OF BEING INTERCONNECTED

- NOTES AND SPECIFICATIONS:
- SIGNS AND LABELS SHALL MEET THE REQUIREMENTS OF THE 2020 ARTICLE 110.21(B), UNLESS SPECIFIC INSTRUCTIONS ARE REQUIRED BY SECTION 690, OR IF REQUESTED BY THE LOCAL AHJ.
 - SIGNS AND LABELS SHALL ADEQUATELY WARN OF HAZARDS USING EFFECTIVE WORDS, COLORS AND SYMBOLS.
 - LABELS SHALL BE PERMANENTLY AFFIXED TO THE EQUIPMENT OR WIRING METHOD AND SHALL NOT BE HAND WRITTEN.
 - LABEL SHALL BE OF SUFFICIENT DURABILITY TO WITHSTAND THE ENVIRONMENT INVOLVED.
 - SIGNS AND LABELS SHALL COMPLY WITH ANSI Z535.4-2011, PRODUCT SAFETY SIGNS AND LABELS, UNLESS OTHERWISE SPECIFIED.
 - DO NOT COVER EXISTING MANUFACTURER LABELS.

CONTRACTOR INFO	DATE	11/27/2025	COMMERCIAL 1234 STREET NAME, CHANUTE, KANSAS 9874 APN: 147852369789456123	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	E-2 SHEET NAME PLACARDS
	PROJECT ID	SP-1						
	CREATED BY	TV						
	CHECKED BY	MSB						
	SCALE	NTS						
	SHEET SIZE	24"X 36"						
ANSI D								
BUILDING OWNER INFO			SHEET INFORMATION					



PHILADELPHIA SOLAR
DELIVERING CLEAN ENERGY SOLUTIONS

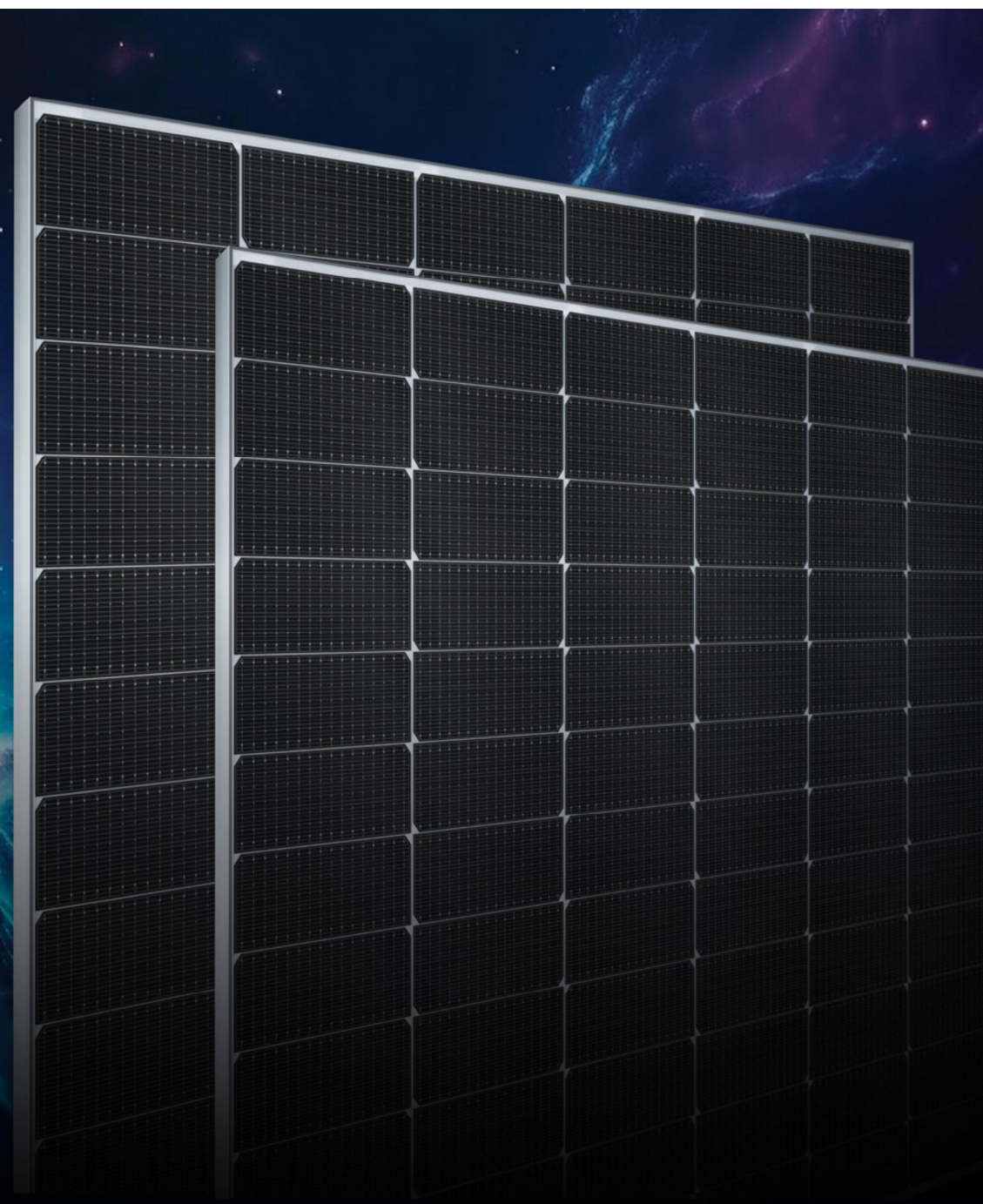
NEXUS

PS-MNB144(HCBF)-xxxW

Half-Cell N-Type 16BB Bifacial Module

595 - 600 Watt

Positive power tolerance of 0 ~+3%



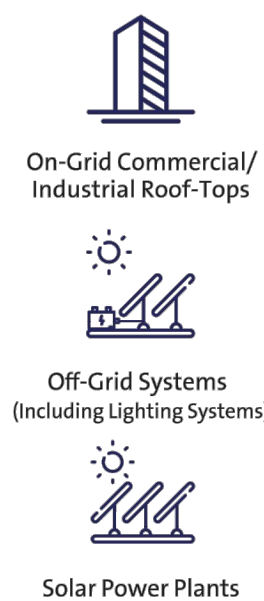
Philadelphia Solar's Mono-Crystal-line N-Type modules with power up to **600Wp** are produced using the state-of-the-art (automated) robotic production lines. These modules are suitable to be used for most electrical power applications and have excellent durability to prevailing weather conditions

CERTIFICATIONS

UL 61215 / UL 61730
IEC 61215 / IEC 61730
CSA C22.2#61730:2019
HALT TEST Highly Accelerated Life And Extended Reliability Test
IEC 61853 PAN File
IEC TS 62804 PID Resistance
IEC 60068 Dust and Sand Resistance
IEC 62716 Ammonia Resistance
IEC 61701 Salt Mist Resistance
Bankability Report
EN ISO 9001: 2015
Quality Management System
EN ISO 14001: 2015
Environmental Management System
EN ISO 45001: 2018
Occupational health and safety management systems



APPLICATIONS



FEATURES

Power output increases by 5-25% from the backside resulting in significantly reduced LCOE and (IRR).

withstand High Mechincal load :
Front (5400 Pascal)
Back (2400 Pascal)

Exceptional Anti-PID performance through the use of optimized mass-production processes and strict materials control.

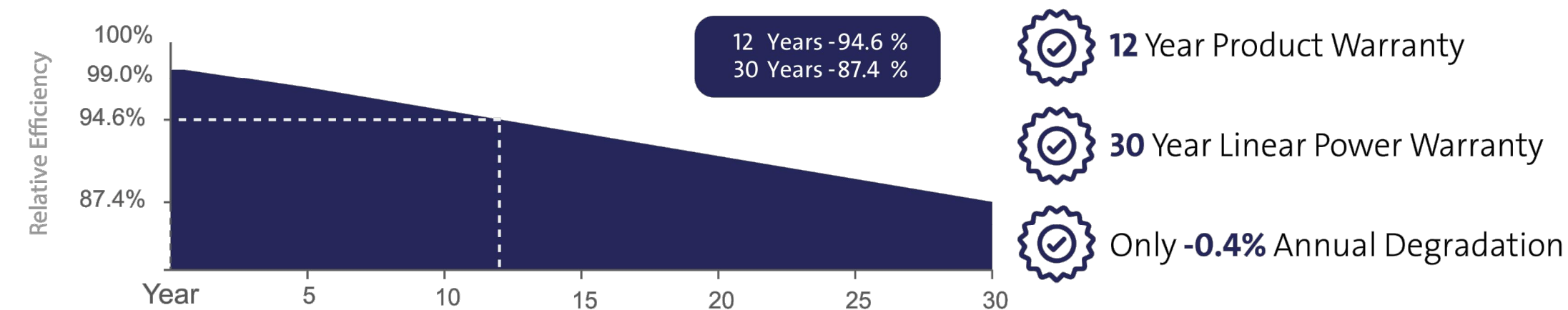
Improved light trapping and current collection technology enhance module power output and reliability.

Less partial shading current mismatch loss so more power output.

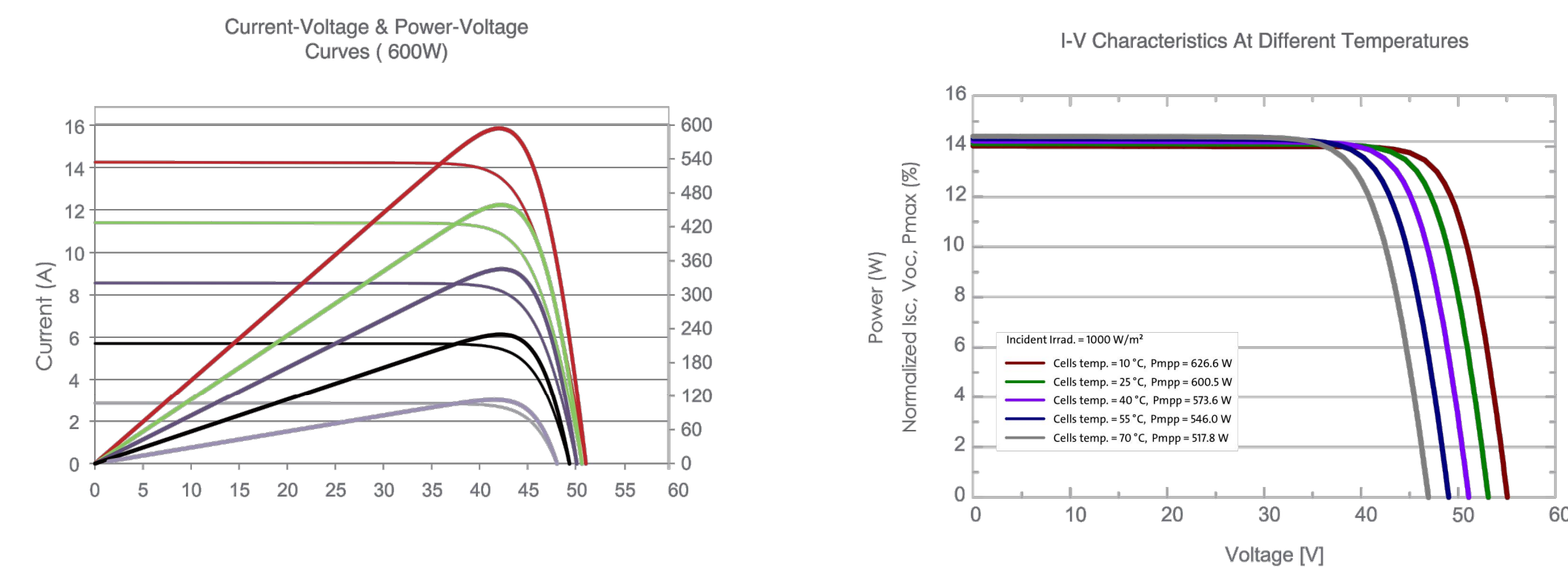
Better temperature coefficients come from half-cell design.



LINEAR PERFORMANCE WARRANTY



Electrical Performance & Temperature Dependence



ELECTRICAL CHARACTERISTICS

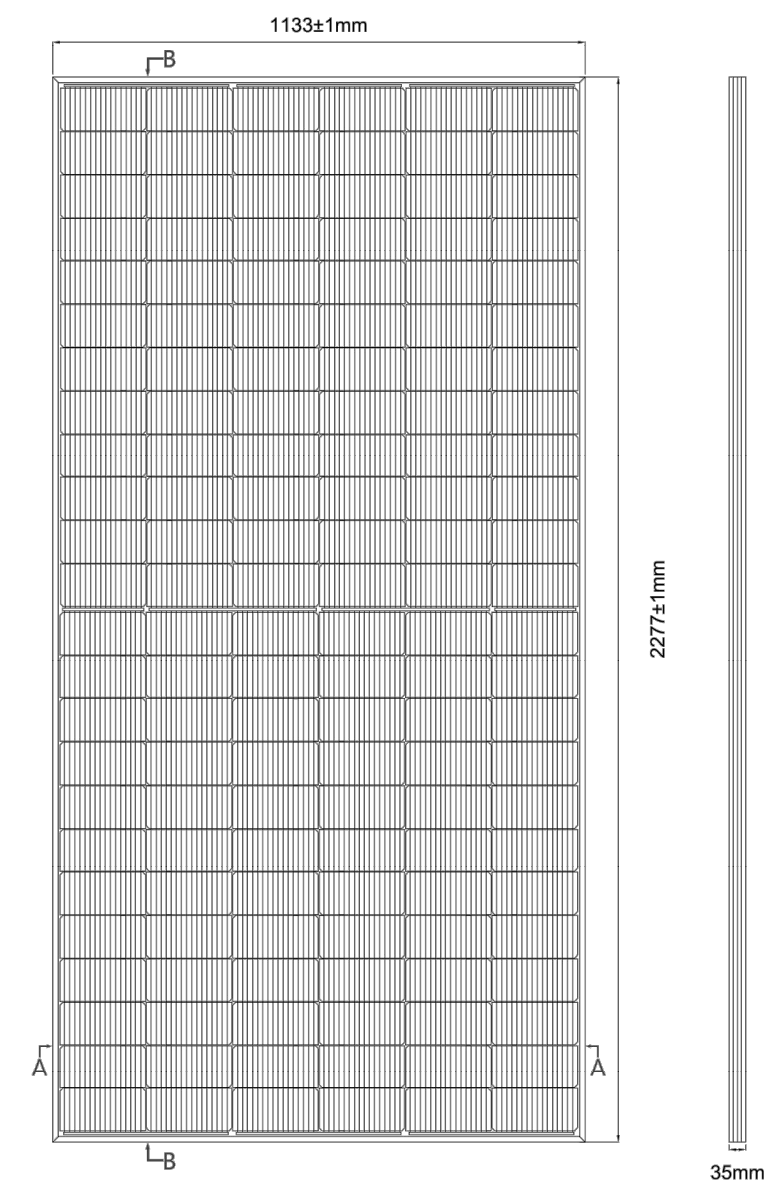
POWER AT STC	595 W	600W
Short Circuit Current - Isc (A)	14.06	14.12
Maximum Power Current - Impp (A)	13.29	13.35
Open Circuit Voltage - Voc (V)	52.91	53.11
Maximum Power Voltage - Vmpp (V)	44.78	44.95
Module Efficiency - η' (%)	23.03%	23.26%
Bifaciality Ratio (%)	80%±5	
Power tolerance (%)	0~+3%	

Values at Standard Test Conditions STC (Air Mass AM 1.5, Irradiance 1000 W/m², Cell Temperature 25° C).

MATERIAL CHARACTERISTICS

Characteristics	Value
Cells per Module	144 (72 x 2)
Cell Type	N Type (TopCon) Mono-Crystalline
Front Surface	3.2mm Tempered AR Coated Glass
Back Cover	Transparent Backsheet
Frame	Anodized Aluminum (Black/Silver)
Junction Box	IP 68 With Original MC4
Cable Length	1200mm Cable length could be customized
Fire Classification	Type 1

MODULE DRAWINGS



THERMAL CHARACTERISTICS

Characteristics	Value
Open Voltage Temperature Coefficient VOC (%/C°)	-0.25
Short Circuit Current Temperature Coefficient ISC (%/C°)	+0.045
Power Temperature Coefficient PMP (%/C°)	-0.29
NOCT (°C)	45±2

PHYSICAL CHARACTERISTICS

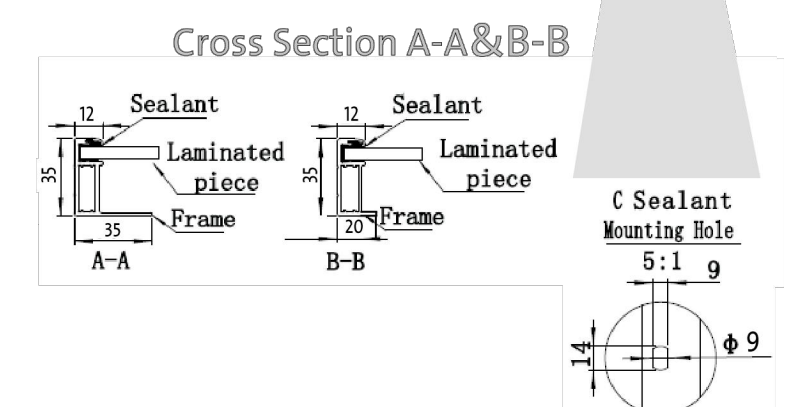
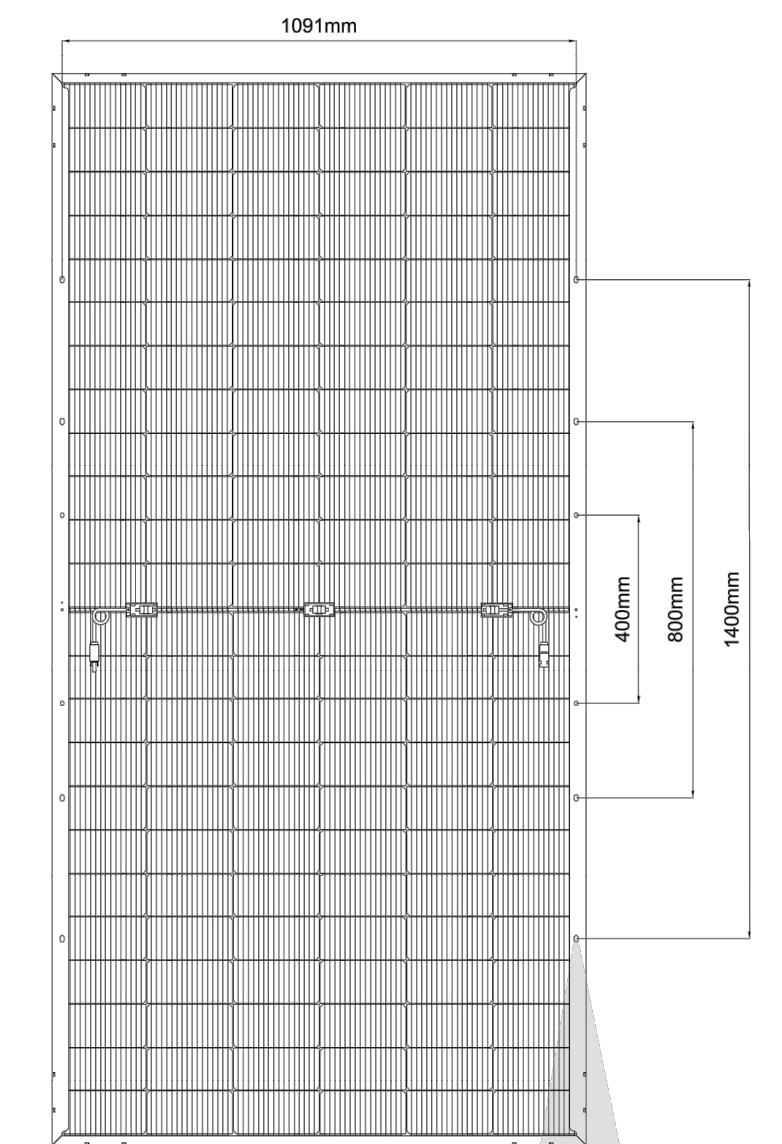
Characteristics	Value
Module Dimensions (mm)	2277 x 1133 x 35
Module Weight (kg)	29 ± 1 Kg
Packaging	Value
Modules per Pallet	31

OPERATING CONDITIONS

Maximum Sytem Voltage - Vmax (V)	1500
Maximum Series Fuse (A)	30
Operating Temperature Range (°C)	IEC: -40 to +85 UL: -40 to +90

Mechanical Load**	Value
Max Static load (Front)	5400 Pa
Max Static load (Back)	2400 Pa
Dynamic load	1000 Pa

- ◆ Tolerance of power Current and Voltage (ISC,VOC)±3 %
- ◆ Datasheet is subjected to change without prior notice, always obtain the most recent version of the datasheet.
- ◆ ** Caution: For professional use only, the installation and handling of PV modules and cleaning modules require professional skills and should only be performed by qualified professionals, please read the Installation and Operation Manual before using the modules, also Cleaning Guidelines



S5-GC(75-125)K-US

Solis Three Phase Grid-Tied Inverters

Efficient

- 8/9/10 MPPTs, max. efficiency 98.8% (CEC efficiency 98.3%)
- > 1.5 DC/AC ratio
- String current up to 16A for higher capacity modules

Economic

- DC side supports "Y" connector

Safe

- Type 4X, C5 Anti-Corrosion Level
- External signal control function
- Nighttime PID recovery function
- AFCI protection, proactively reduces fire risk
- High quality components from globally recognized suppliers

Smart

- Intelligent string monitoring, smart I-V curve scan
- Remote firmware upgrade with simple operation⁽¹⁾

Models:

S5-GC75K-US / S5-GC80K-US
S5-GC90K-US / S5-GC100K-US
S5-GC125K-US

Ordering: S5-GC(75-125)K-US

- APST (APS MLRSD Transmitter)
- RSS (Tigo MLRSD Transmitter)
- NEPT (NEP MLRSD Transmitter)



(1) Requires the user to use Solis monitoring

DATASHEET

S5-GC(75-125)K-US

Models	75K	80K	90K	100K	125K
Input DC					
Max. input voltage	1000 V				
Rated voltage	600 V				720 V
Start-up voltage	195 V				
MPPT voltage range	180 - 1000 V				
Max. input current	8 × 32 A	9 × 32 A	10 × 32 A		
Max. short circuit current	8 × 50 A	9 × 50 A	10 × 50 A		
MPPT number / Max. input strings number	8 / 16	9 / 18	10 / 20		
Output AC					
Rated output power	75 kW	80 kW	90 kW	100 kW	125 kW
Max. apparent output power	75 kVA	80 kVA	90 kVA	100 kVA	125 kVA
Max. output power	75 kW	80 kW	90 kW	100 kW	125 kW
Rated grid voltage	3Φ/PE, 480 V				
Rated grid frequency	60 Hz				
Max. output current	90.2 A	96.2 A	108.3 A	120.3 A	150.4 A
Power Factor	> 0.99 (0.8 leading - 0.8 lagging)				
THDi	< 3%				
Efficiency					
Max. efficiency	98.7%		98.8%		
CEC efficiency	98.3%		98.2%		
Protection					
DC reverse-polarity protection	Yes				
Surge protection	DC Type II / AC Type II				
Ground fault monitoring	Yes				
Anti-islanding protection	Yes				
Strings monitoring	Yes				
I/V Curve scanning	Yes				
Rapid shutdown	Yes				
Integrated AFCI	Yes				
Integrated PID recovery	Yes				
AC switch	Yes				
General Data					
Dimensions (W × H × D)	41.9 × 22.3 × 13.6 in (1065 × 567 × 344.5 mm)				
Weight	187 lbs (85 kgs)				200 lbs (91 kgs)
Topology	Transformerless				
Self consumption (night)	< 2 W				
Relative humidity	0 - 100%				
Operating ambient temperature range	-22°F to 140°F (-30°C to 60°C)				
Storage environment	-40°F to 176°F (-40°C to 80°C)				
Ingress protection	TYPE 4X				
Cooling concept	Intelligent fan-cooling				
Max. operation altitude	13,120 ft (4000 m)				
Compliance	UL1741SB, IEEE 1547-2018, UL1699B, UL1998, FCC Part15 ClassB, California Rule 21, Heco Rule 14H, NEC 690.12-2020, CAN/CSA C22.2107.1-1				
Features					
DC connection	MC4 connector				
AC connection	OT Terminal (max. 350 MCM)				
Display	LCD				
Communication	RS485, Optional: Wi-Fi, Cellular				

Ordering guidelines: Determine the basic model and add your desired features from above.
Ex: S5-GC75K-US-APST (Inverter with APS transmitter)



Raising the bar in innovative
DC MLPE solar power systems



RSD-S-PLC

- Meets NEC 2017 & 2020 (690.12) requirements
- Executes rapid shutdown of system when Transmitter-PLC signal is absent
- Meets SunSpec requirements

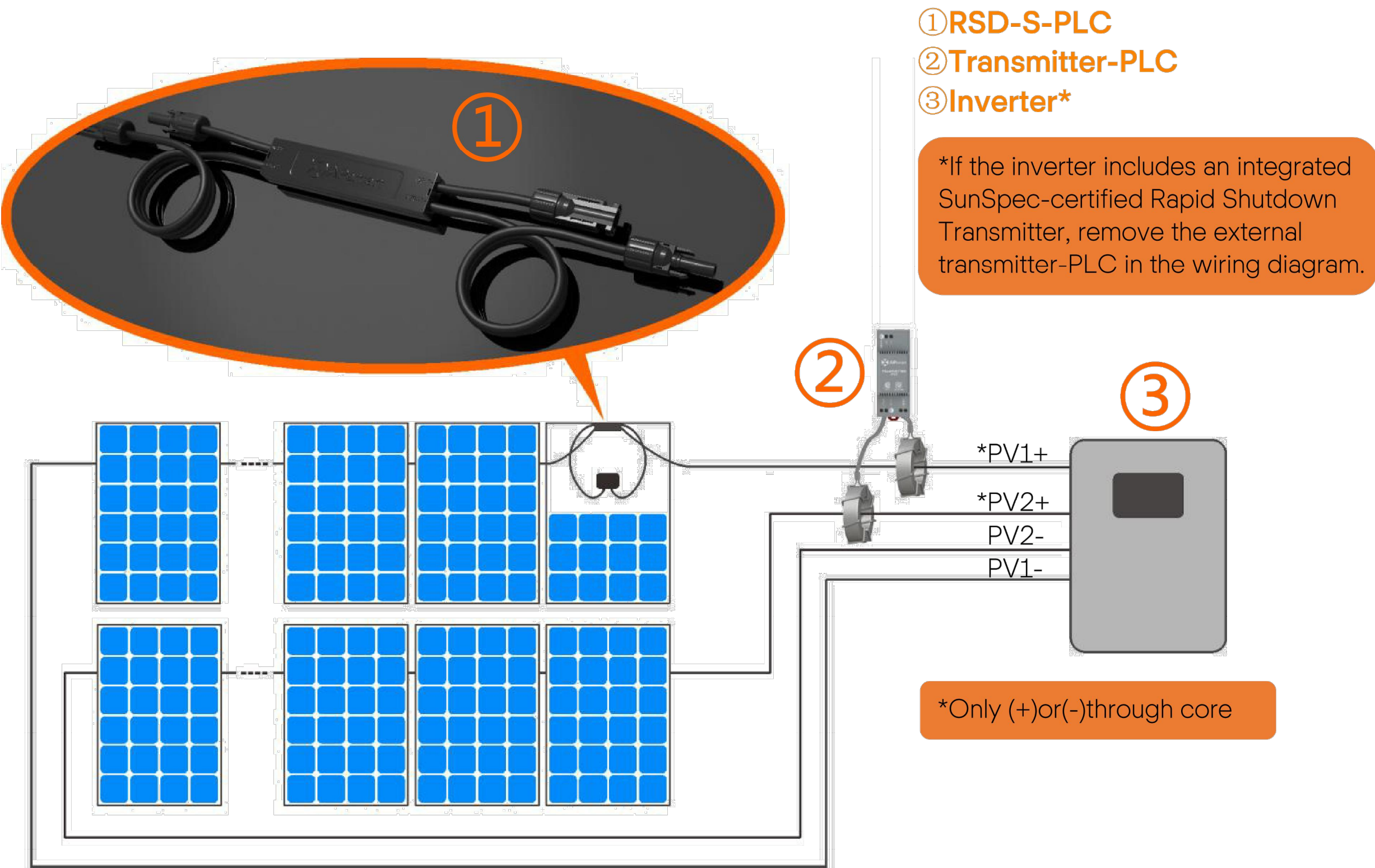
The RSD-S-PLC meets SunSpec requirements,maintaining normal function by continually receiving a heartbeat signal from the APsmart Transmitter. The RSD executes rapid system shutdown when the Transmitter signal is absent. Users can manually execute rapid shutdown using Transmitter breaker switch.

RSD-S-PLC TECHNICAL DATA

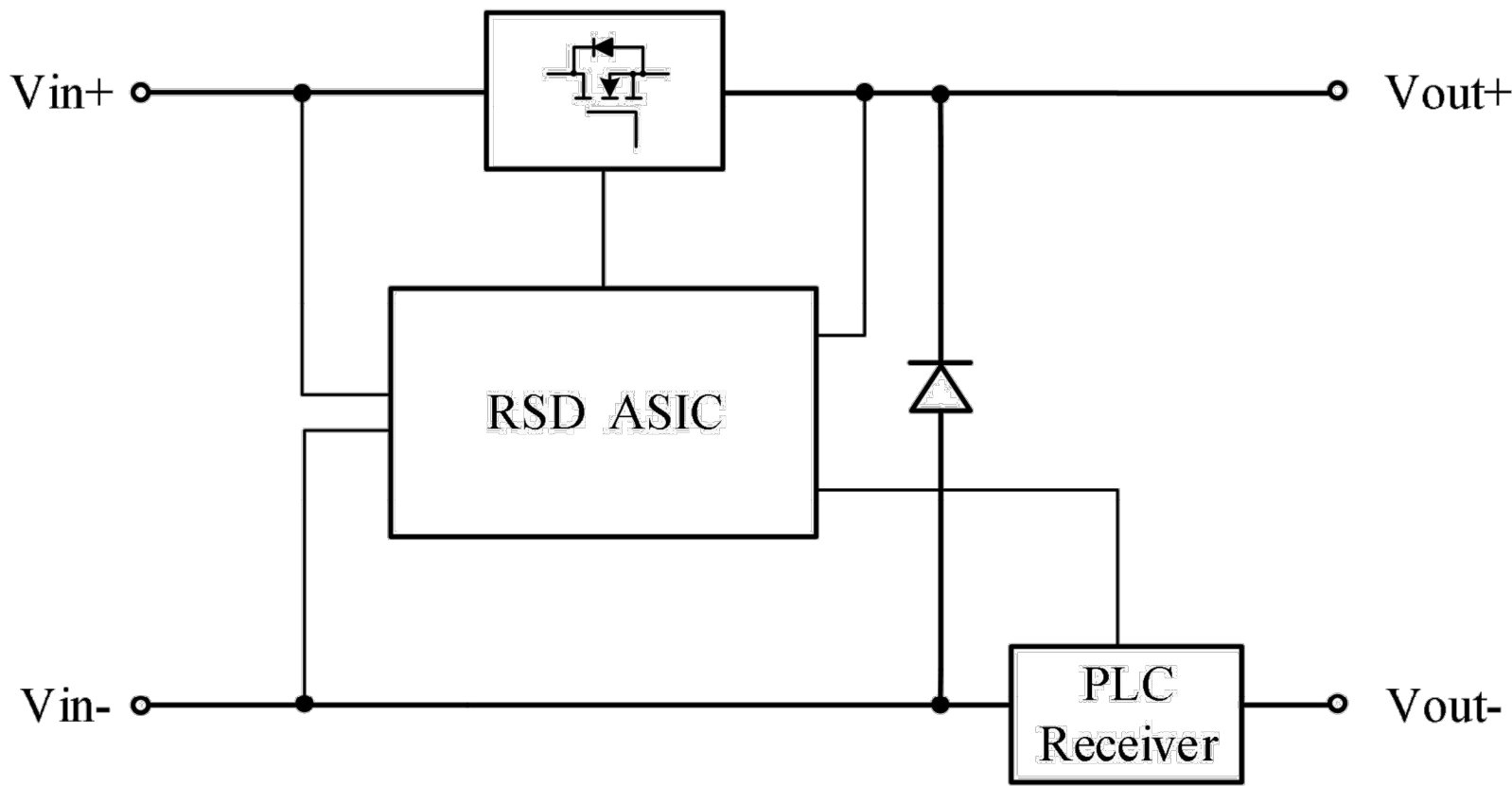
MODEL	RSD-S-PLC
INPUT DATA (DC)	
Input Operating Voltage Range	8-80V
Maximum Cont. Input Current (Imax)	15A
Maximum Short Circuit Current (Isc)	25A
OUTPUT DATA (DC)	
Output Operating Voltage Range	8-80V
Maximum System Voltage	1000V/1500V
Maximum Series Fuse Rating	30A
MECHANICAL DATA	
Operating Ambient Temperature Range	-40°F to +185°F (-40 °C to + 85 °C)
Dimensions (without cable & connectors)	5" x 1.2" x 0.6"(129 mm x 30 mm x 16 mm)
Cable Length	Input 250mm/Output 1200mm
Cable Cross Section Size	UL:12AWG
Connector	Input: Stäubli MC4 PV-KBT4&KST4 or Customized Output: APsystems specified or Customized
Enclosure Rating	NEMA Type 6P/IP68
Protection Temperature	100°C
FEATURES & COMPLIANCE	
Communication	PLC
Safety Compliance	NEC 2017 & 2020 (690.12); UL1741; CSA C22.2 No. 330-17; IEC/EN62109-1; 2PFG2305 FCC Part15; ICES-003;IEC/EN61000-6-1/-2/-3/-4



RSD-S-PLC WIRING DIAGRAM



WORKING SCHEMATIC DIAGRAM



ORDERING INFORMATION

415002	1500V UL, 1.2m cable, Stäubli MC4 PV-KBT4&KST4
415001	1000V UL, 1.2m cable, Customized connector



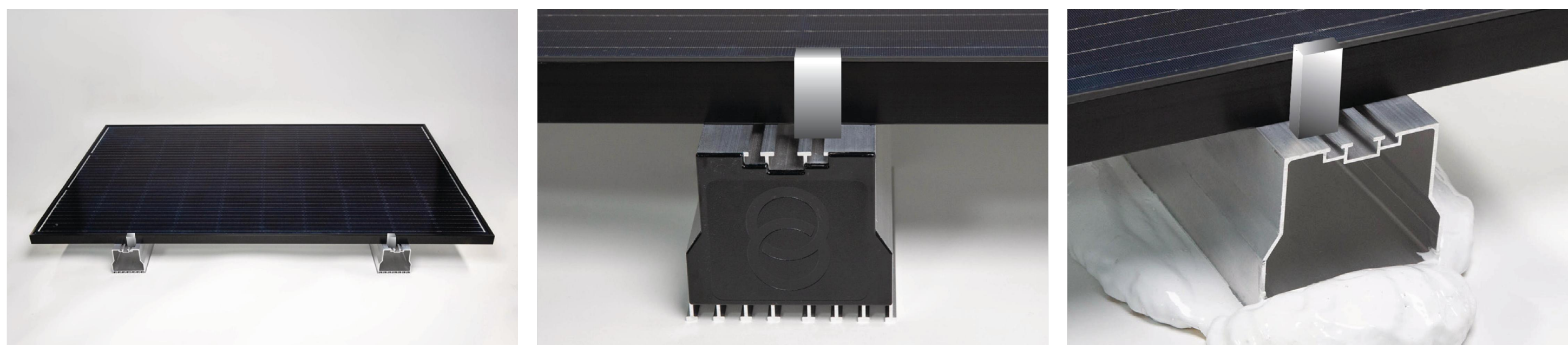
SOLARSTACK[®] 101

FOR CONTRACTORS

MIAMI-DADE COUNTY
APPROVED
NOA No. 21-0520.05

► What is Solar Stack?

Solar Stack is the only solar panel mounting system that avoids all roof penetrations and damage while meeting the strictest building codes in North America.



► How does it work?

Solar Stack is bonded to the roof membrane with spray polyurethane foam adhesive.



► Products



8" flush mounts (SS-8)

12" flush mounts (SS-12)

Tilt Mounts, 5° and 10° options
(SS-8 TK5 & SS-8 TK10)

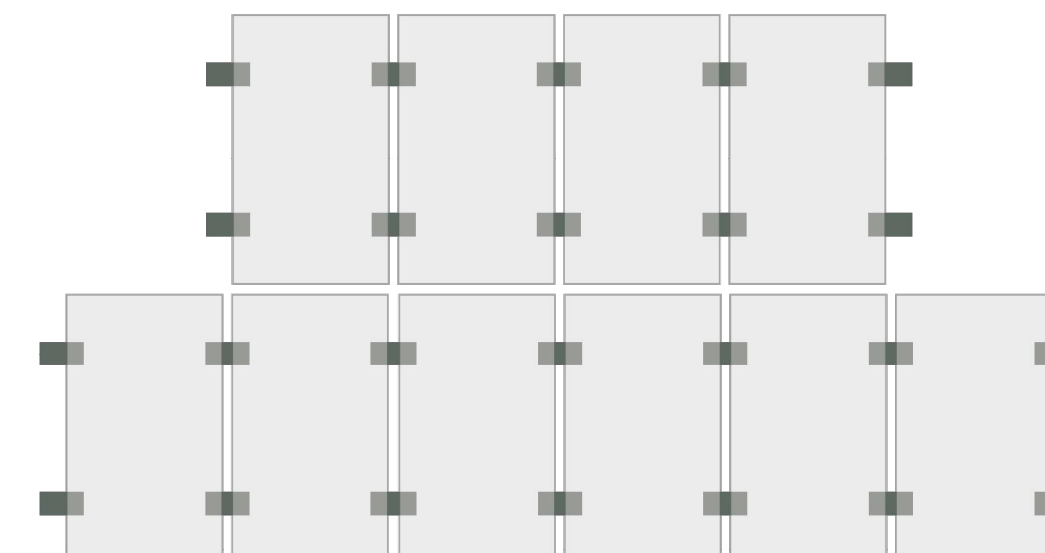
► How many Solar Stacks per panel?

Solar Stacks share in-between panels

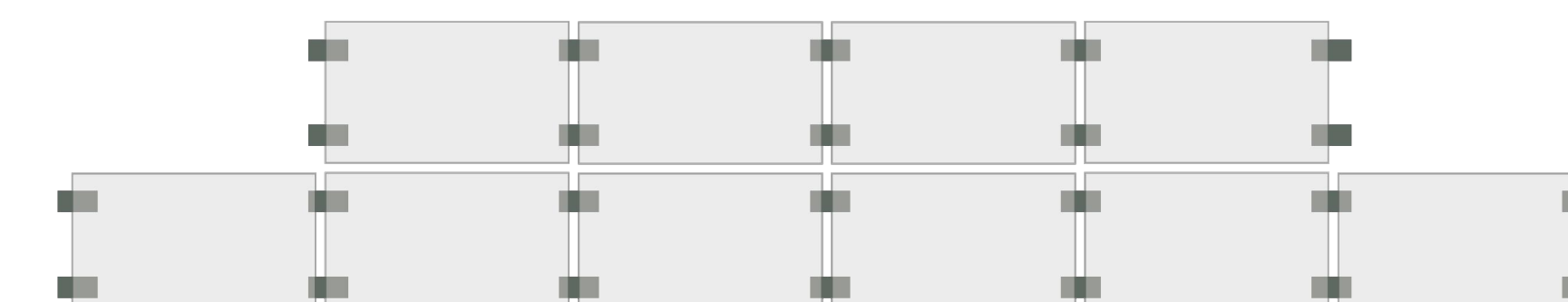
To find the exact amount of SS needed, take the amount of panels x 2 + the amount of rows x 2 = number of Solar Stacks required.

$$\begin{array}{rcl} \text{amount of panels} & 10 & \times 2 & 20 \\ + & & & \\ \text{the amount of rows} & 2 & \times 2 & 4 \\ = & & & \\ \text{Solar Stacks required} & & & 24 \end{array}$$

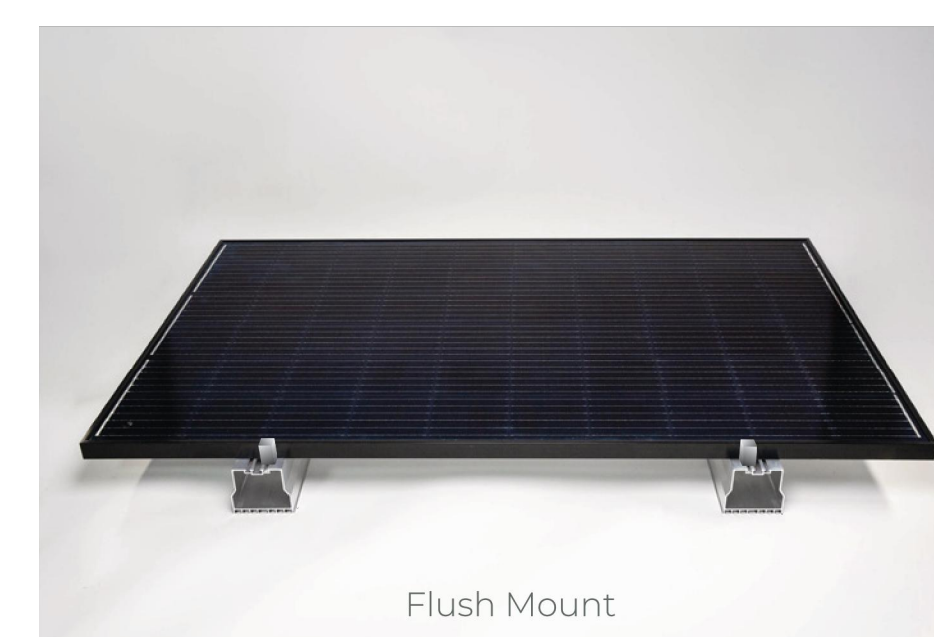
NOTE: Calculation is same for either orientation.



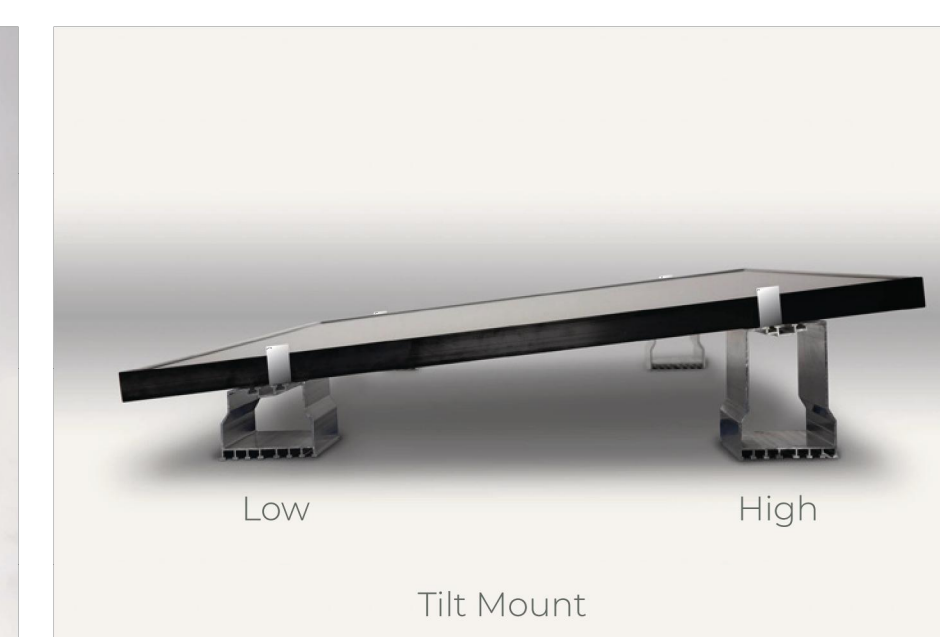
Flush Mount (Portrait)



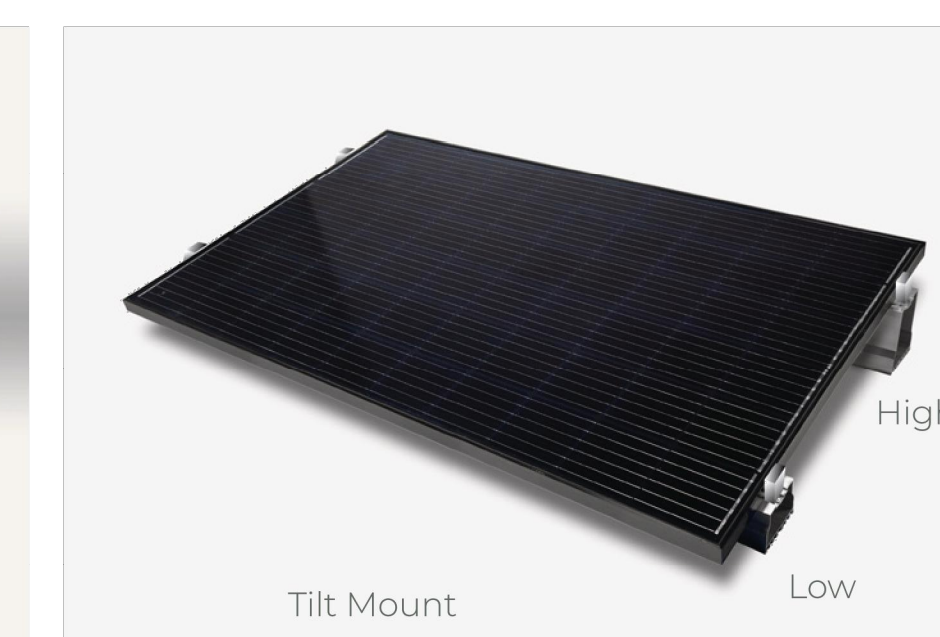
Tilt Mount (Landscape)



Flush Mount



Tilt Mount



Tilt Mount

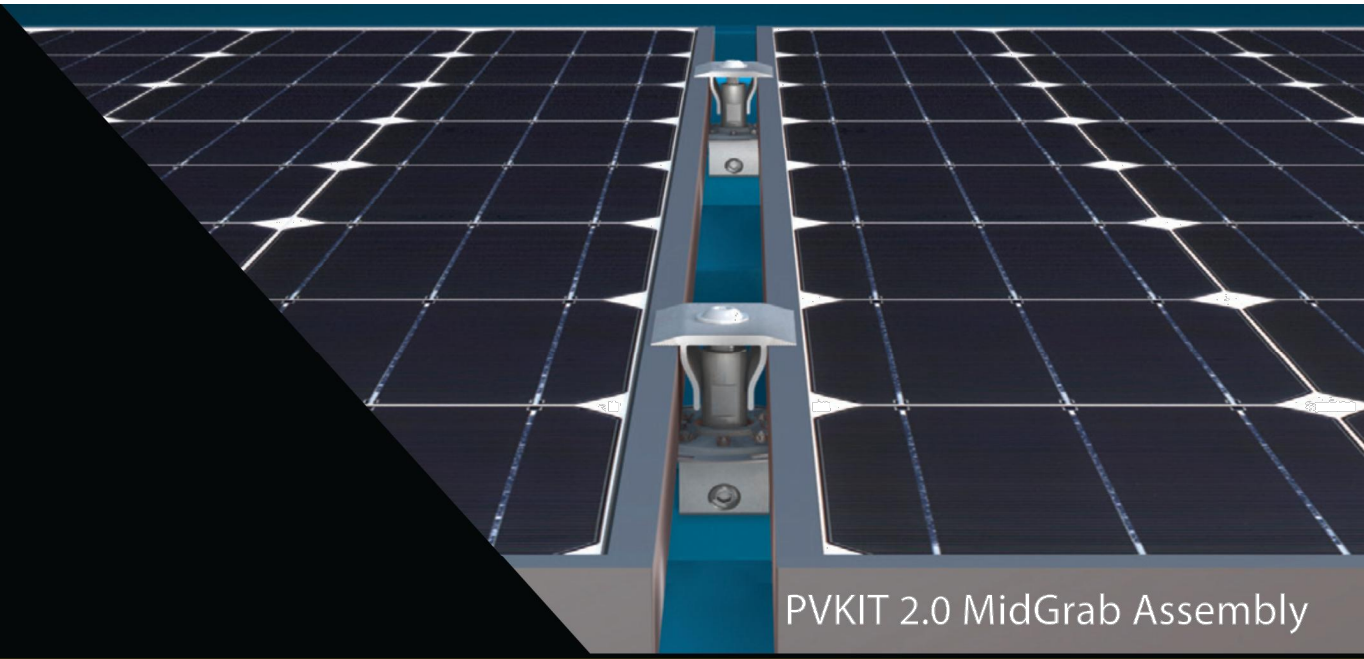


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PVKIT® 2.0: New Design



The PVKIT 2.0 is furnished with the hardware shown at right, excluding the attachment clamp, which is supplied separately. The PVKIT 2.0 is compatible with most common metal roofing materials, including copper.

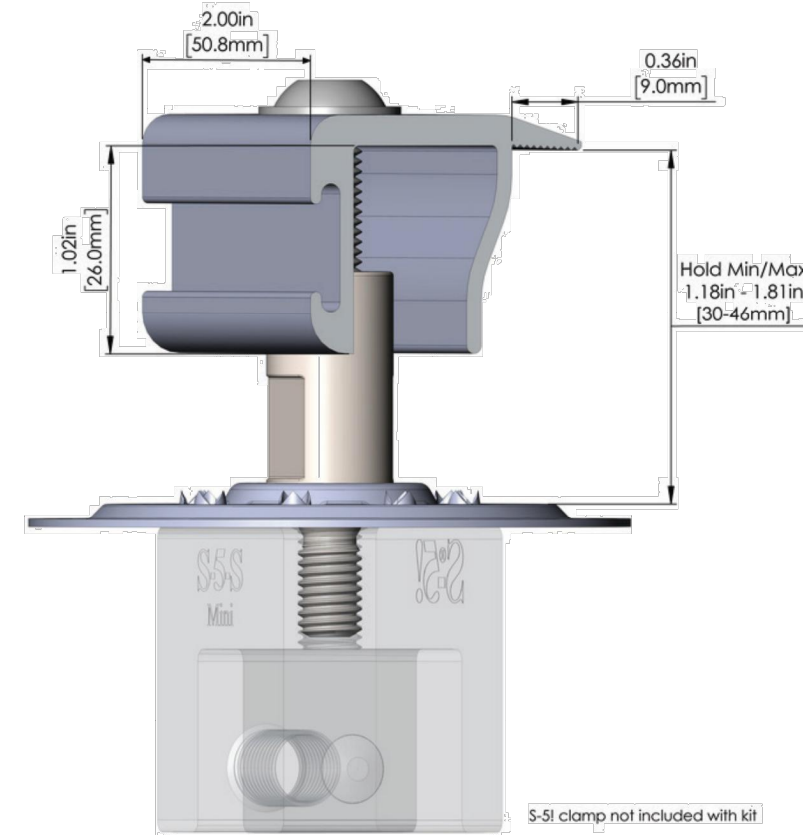
The Module Placement Bevel Guide makes the module placement easier. The mounting disk is multi-directional and rails are not required. The PV grab ears, holding the solar panels in place, are broader to allow for ease of installation and precise module engagement.

Accommodating module thicknesses between 30 and 46mm, the PVKIT 2.0 fits the majority of solar panels on the market. Using the S-5! mini clamps, it fits most standing seam metal roofs. When paired with other S-5! products, the PVKIT 2.0 and EdgeGrab or MidGrab will also work on most exposed fastener roofs including corrugated metal roofs. The MidGrab is designed to fit mid conditions (two adjacent panels), while the new EdgeGrab is designed specifically for end conditions.

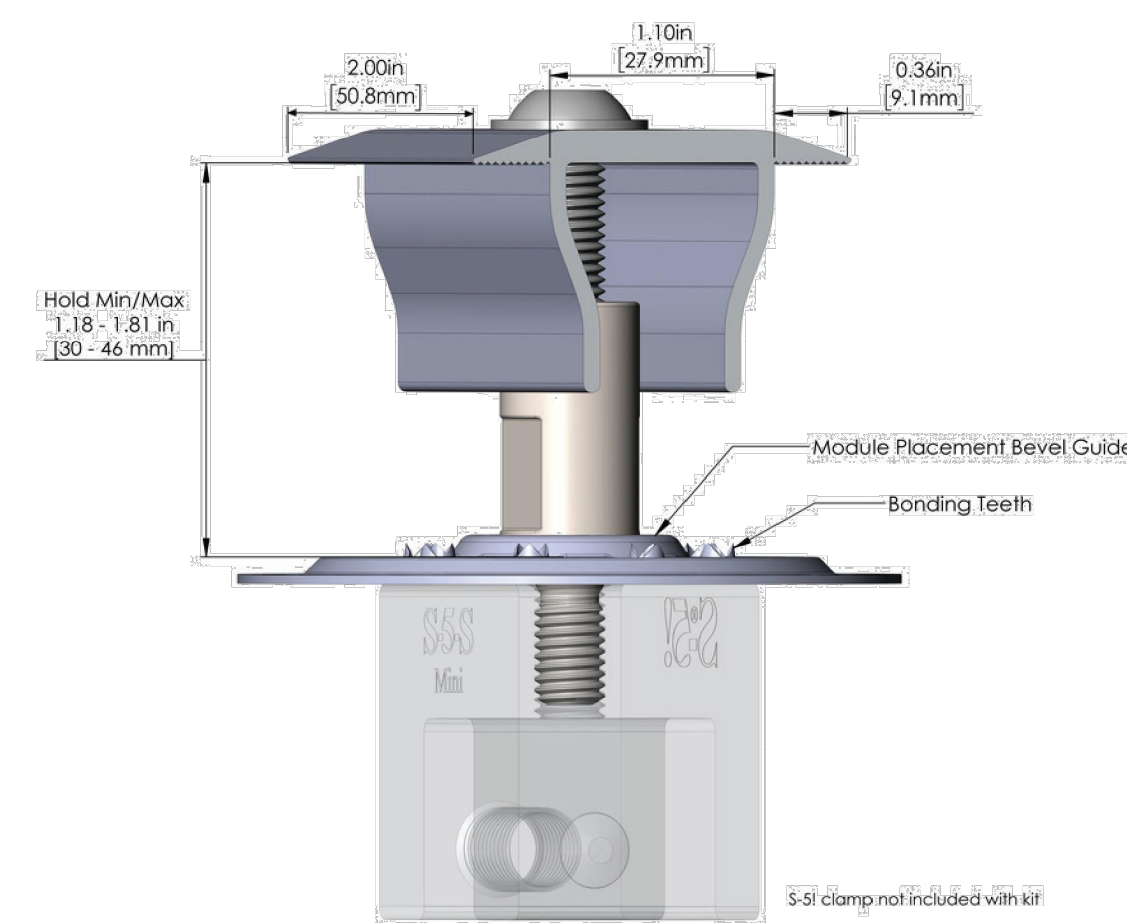
Wind dynamics are complex; thus, each system should be reviewed by a qualified licensed professional who understands wind effects prior to purchase and installation. For more detailed information including specifications, installation instructions, and CAD drawings, visit www.S-5.com or your PVKIT 2.0 distributor.

The PVKIT 2.0 continues to be the easiest, most cost-effective way to install solar panels directly to standing seam and exposed fastener metal roofs, remaining the most popular choice worldwide.

PVKIT 2.0 EdgeGrab



PVKIT 2.0 MidGrab



Certain components featured in illustration may not be UL listed. Due to the variety of attachment needs, S-5-PVKIT 2.0 are sold separately from S-5! clamps.

Distributed by:

S-5!® Warning! Please use this product responsibly!

The independent lab test data found at www.S-5.com can be used for load-critical designs and applications.

Products are protected by multiple U.S. and foreign patents. For published data regarding holding strength, fastener torque, patents, and trademarks, visit the S-5! website at www.S-5.com. Copyright 2021, Metal Roof Innovations, Ltd. S-5! products are patent protected.

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The concept of combining photovoltaic arrays with standing seam metal roofing is growing—and for good reasons. A standing seam metal roof has a life expectancy consistent with that of framed PV modules—a 30-year power source on a 40-year roof, along with zero-penetration technology, creates the most sustainable roof system available with alternative power generation, all without compromising the roof manufacturer's warranty! PVKIT® 2.0 is also a great solution for attaching PV modules directly to many exposed fastener roofs when paired with S-5! brackets.

S-5! has introduced a new and improved PVKIT, boasting an improved installation experience for PV mounting technology. The kit comes preassembled with either the MidGrab or EdgeGrab for easier and more efficient installation. The kits were designed with thread lock on the standoff bolt so that the grab will seat to the PV Module frame by using one tool to drive the top bolt, eliminating a step required in the previous PVKIT. The PVKIT 2.0 features bonding teeth, which are more aggressive to secure a better ground path. No lugs or wire required except to connect one string of modules to another and to ground the system.

The S-5 PVKIT 2.0 is built to save you time and money —
The Right Way® to install solar to your metal roof.

PVKIT 2.0 Features:

Pre-assembled kit saves time and money

Only *one tool* needed for installation

Bolt head uses standard hex bit tip which is provided

Improved single piece EdgeGrab installs with ease

Low profile bolt head provides a sleek and clean finish

Also available in black by special order only

MidGrab leaves 1" gap between modules, allowing reduction per ASCE7

UL 2703 Listed



PVKIT 2.0 EdgeGrab Assembly

PVKIT® 2.0 MidGrab or EdgeGrab



888-825-3432 | www.S-5.com

S-5!®

The Right Way!®

S-5-U Clamp

The S-5-U clamp is by far our most popular and most versatile clamp. It fits about 85% of the standing seam profiles manufactured in North America—including most structural and architectural profiles. It can be used on vertically oriented seams and, by rotating the clamp 90 degrees, it can also be used on most horizontal 2" seam profiles.

Its simple design, generous dimensioning, and multiple hole orientations are what make the S-5-U clamp so versatile for use with the S-5!® snow retention products, such as ColorGard®, as well as with other heavy-duty applications.

Installation is as simple as setting the specially patented round-point setscrews into the clamp, placing the clamp on the seam, and tightening them to the specified tension. Then, affix ancillary items using the bolt provided with the product. Go to www.S-5.com/tools for information and tools available for properly attaching and tensioning S-5! clamps.

S-5-U Mini Clamp

The S-5-U Mini is a bit shorter than the S-5-U and has one setscrew rather than two. The mini is the choice for attaching all kinds of rooftop accessories: signs, walkways, satellite dishes, antennas, rooftop lighting, lightning protection systems, solar arrays, exhaust stack bracing, conduit, condensate lines, mechanical equipment—just about anything!*

*S-5! mini clamps are not compatible with, and should not be used with S-5! SnoRail™/SnoFence™ or ColorGard® snow retention systems.



The S-5-U clamp is our most popular and versatile clamp, fitting about 85% of the standing seam profiles in North America.

S-5-U and S-5-U Mini



888-825-3432 | www.S-5.com

S-5!®

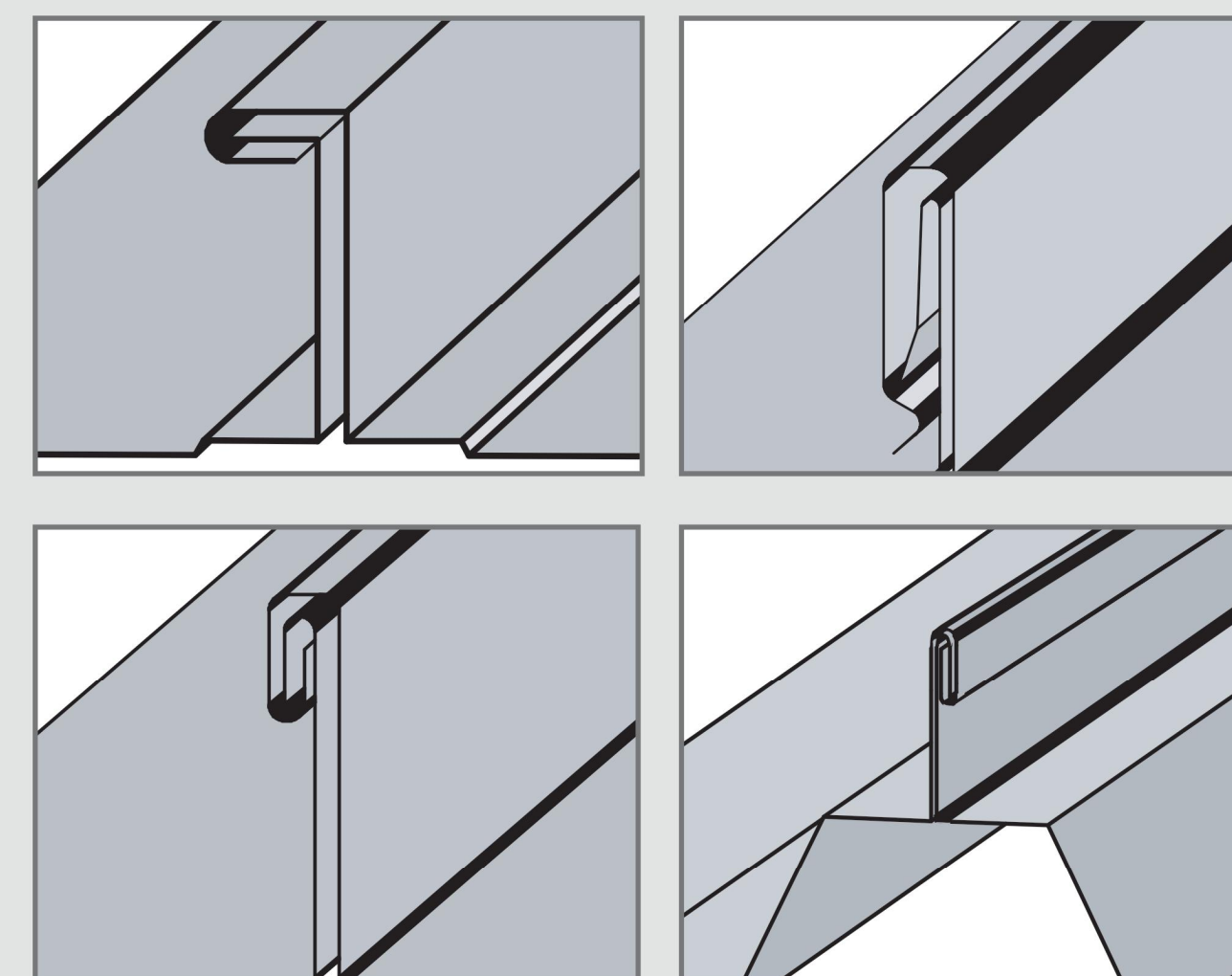
The Right Way!®

The strength of the S-5-U clamp is in its simple design. The patented setscrews will slightly dimple the metal seam material but not pierce it—leaving the roof manufacturer's warranty intact.

The **S-5-U and S-5-U Mini clamps** are each furnished with the hardware shown to the right. Each box also includes a bit tip for tightening setscrews using an electric screw gun. A structural aluminum attachment clamp, the S-5-U is compatible with most common metal roofing materials excluding copper. All included hardware is stainless steel. Please visit www.S-5.com for more information including CAD details, metallurgical compatibilities and specifications.

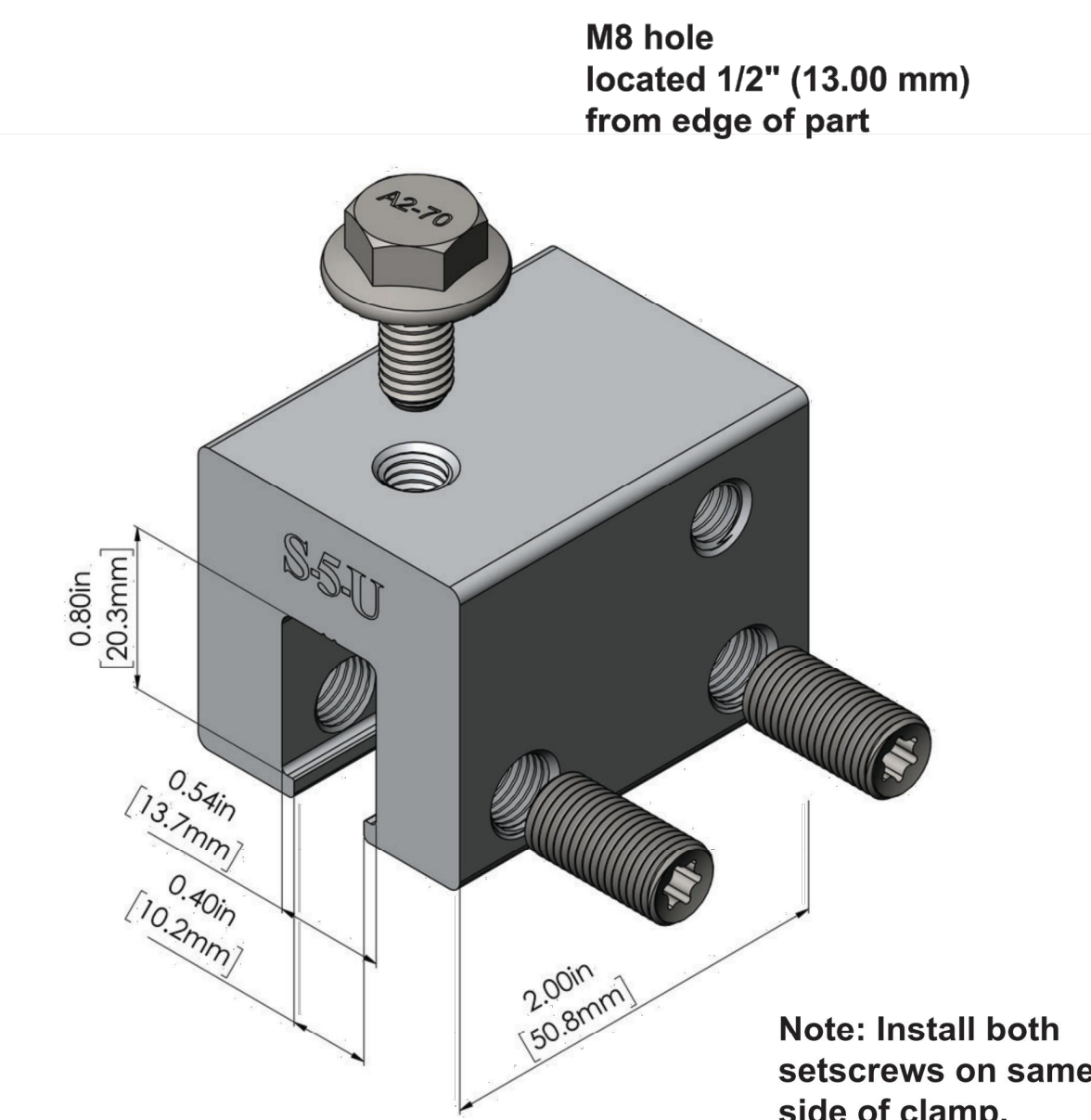
The S-5-U clamp has been tested for load-to-failure results on most major brands and profiles of standing seam roofing. The independent lab test data found at www.S-5.com can be used for load-critical designs and applications. S-5!® holding strength is unmatched in the industry.

Example Profiles

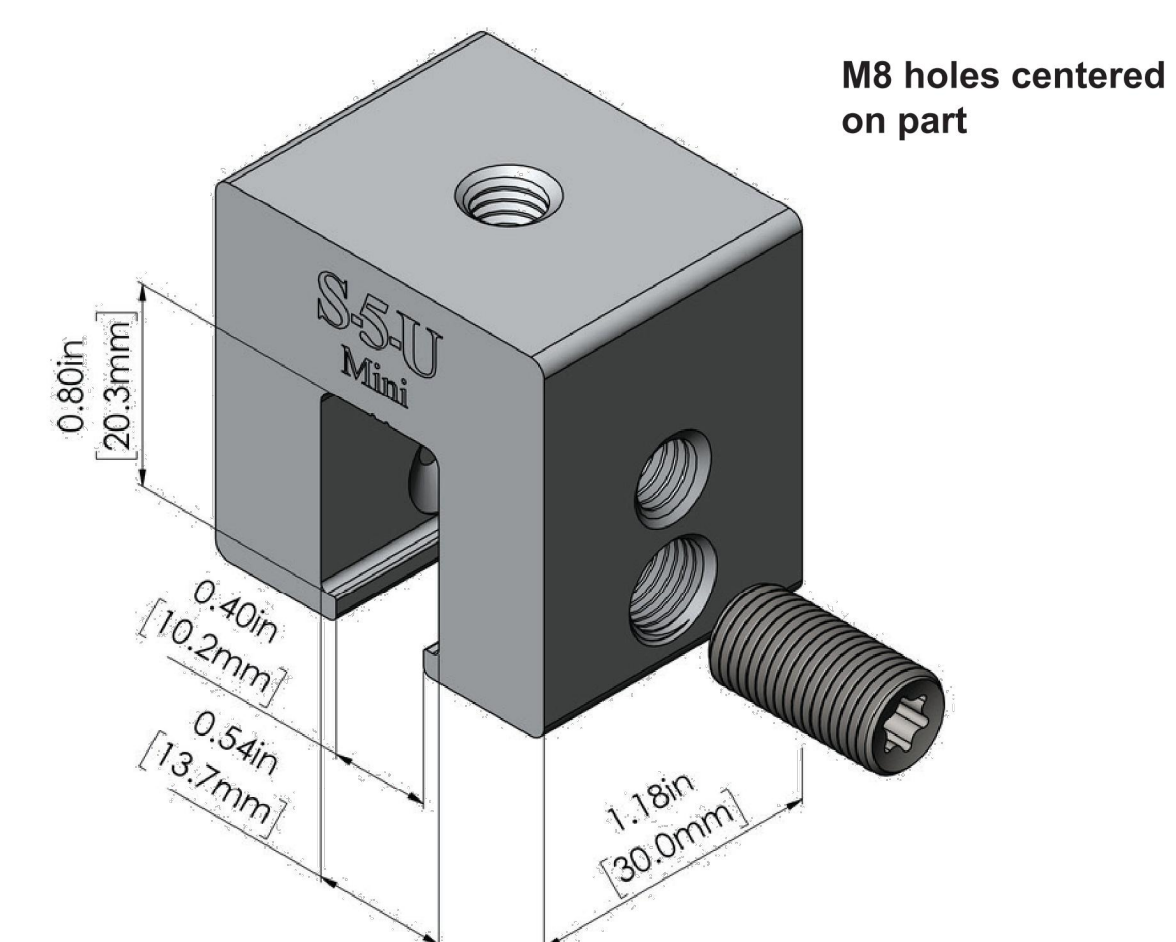


For horizontal seams under 0.65", do not use this clamp. Visit www.S-5.com for more detailed information and proper clamp usage.

S-5-U Clamp



S-5-U Mini Clamp



S-5!® Warning! Please use this product responsibly!

Products are protected by multiple U.S. and foreign patents. Visit the website at www.S-5.com for complete information on patents and trademarks. For maximum holding strength, setscrews should be tensioned and re-tensioned as the seam material compresses. Clamp setscrew tension should be verified using a calibrated torque wrench between 160 and 180 inch pounds when used on 22ga steel, and between 130 and 150 inch pounds for all other metals and thinner gauges of steel. Consult the S-5! website at www.S-5.com for published data regarding holding strength.

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